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Potential field-based formation tracking control for multi-UGV system with detection behavior and collision avoidance[☆]

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Abstract

This paper investigates the time-varying formation tracking and collision avoidance problems for multi-unmanned ground vehicle (UGV) system which consists of omnidirectional vehicles on directed topology. To prevent the failure of collision avoidance caused by the lack of information interaction when the directed graph is not unilaterally connected, partial vehicles are elected as observers and their detection behaviors represented by sectors are realized by potential field and yaw control. Firstly, an improved repulsive potential field force is designed, and two composite potential field forces are proposed for single-to-single detection and single-to-multiple detection situations. Inspired by the equilibrium of forces, it is proven that the reconstructed potential field force which is not affected by the complexity of the collision situation can eradicate the local minima or deadlock states. Secondly, in order to generate detection behaviors according to a basic communication graph, an algorithm is put forward. Then, the protocol of position and velocity control, as well as the protocol of yaw control, are presented respectively. The former ensures the realization of formation tracking, while the latter, together with composite potential field forces, makes the detection behavior continuous. Moreover, according to Lyapunov-based

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theory, it is proven that the given method can achieve time-varying formation tracking and collision avoidance. Finally, simulations demonstrate the theoretical results.

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1. Introduction

To date, the control of multi-agent systems has been thoroughly discussed, including consensus control [1,2], formation control [3–5], formation tracking control [6,7], formation-containment control [8,9], etc. For large-scale multi-agent systems with complex communication networks, information interactions between individuals are easily influenced and unexpected collisions may occur. Therefore, as the exploring of this field becomes increasingly comprehensive, safety topic for multi-agent systems which mainly includes information and physical safety has attracted much attention [10–13].

As one of the bases for agents to complete the mission, physical safety mainly refers to collision avoidance, which can be fulfilled by artificial potential field (APF) [14], path planning-based method [15–17], control barrier functions [18,19], learning-based method [20,21], etc. Among them, the APF-based method has been commonly utilized in practical applications for a long time because of its simplicity. Finite-time consensus and collision avoidance for multi-agent systems were investigated in [22]. Except for the leaderless case, the authors developed distributed finite-time observers to estimate the leader's velocity in the leader-follower situation. Then a finite-time APF-based position consensus algorithm was proposed to ensure velocity matching, consensus boundedness, collision avoidance, and connectivity maintenance. To realize real-time trajectory planning for unmanned surface vehicles, global paths were generated as references to avoid local minima of trajectory in [23]. Attractive APF was used to guide the agent along its global path, two different repulsive APFs were utilized to keep the agents away from shores and avoid collisions between agents. [24] focused on the APF-based method for second-order nonlinear multi-agent systems, where a neural network was used to solve the problems of unknown nonlinear dynamics. Under communication constraints, the topic for formation tracking with collision avoidance was dealt with by [25]. By continuous-discrete time observers, an output feedback formation controller was designed under directed topology. Then APF was utilized to ensure collision avoidance. Based on undirected topology, a distributed fixed-time sliding manifold as well as a repulsive potential function was proposed with consideration of external disturbances in [26]. A novel sliding manifold which evolved from APF was proposed in [27], and an event-triggered strategy was applied to the distributed formation control of leader-follower systems. The neural-based observer was used to estimate both the velocity and uncertainty in [28]. To approximate the unknown derivatives of the formation reference, a sliding mode differentiator was also introduced. Finally, by applying APFs, a sliding mode controller was constructed. Though the APF-based strategy is widely exploited, however, it is increasingly not favored by scholars for annoying conundrums (e.g., deadlock dilemma) caused by its own characteristics, and in most of the existing literature, these fatal defects cannot be systematically improved. For instance, although the convergence of APF-based control protocol was verified, the discussion about local minima was not sufficient in [22]. Similar problem appears in [24–27].

For APF-based collision avoidance, each control objective is usually described as an independent part. The whole control function is generated by summing up all control parts, which results in the deadlock state or local minima when the control effects of different parts are contradictory. According to the generation mechanism of local minima, some scholars have made improvements [29–33]. For high-order multi-agent systems, the output formation tracking problems with collision avoidance were investigated in [29]. With the uncertainties of dynamics for agents and the unknown control input of the leader, the outputs of followers were capable to track the output of the leader. To avoid local minima, a temporary reference points strategy was introduced. Specifically, a temporary target point was generated near the obstacle to guide the agent to move away from the deadlock state. An orthogonal vector field method was proposed in [30] to make the overall repulsive vector orthogonal to the attractive vector, which improved the efficiency of collision avoidance. The method could solve the deadlock problem caused by repulsive and attractive vectors. In [31], distributed control protocol for a multi-agent system was divided into three objectives: reaching the goal, circulating the target, and avoiding collisions. Based on cross-product, the authors proved three gradient fields were orthogonal to each other, which could solve the deadlock problem. Formation control problem with collision avoidance was considered in [32,33], where repulsive vector field (RVF)-based method was utilized to eliminate local minima. Specifically, the potential field was considered as a counterclockwise unstable focus centered at the position of the agent. By reconstructing the potential field function, the undesired saddle point (local minimum) in the region of the repulsive potential field was removed. The literature above eliminates local minima by different methods, however, some flaws exist. In particular, the individual dynamics and the application scenarios were relatively simple, and collision might not be avoided when multiple agents were in some special positions [30,31]. Besides, the distribution of repulsive vectors would become complicated and several undesired saddle points may exist in phase portrait when an agent is in multiple repulsive potential fields at the same time, which makes it complex to analyze and design potential field function [32,33]. Similarly, the selection of a temporary state is also susceptible to the complexity of the collision situation, which was not discussed in [29].

In addition, although formation control can be achieved on the directed graph which contains at least one spanning tree, most of the research on APF-based algorithms is developed for undirected and directed graphs, which are unilaterally connected. On the one hand, the process of Lyapunov-based stability analysis for the system described by the undirected graph will not be affected by the introduction of APF as shown in [22,24]. On the other hand, the basic requirement for position information in collision avoidance may be fulfilled in the directed graph which is unilaterally connected when not considering the use of additional operations to obtain position information [28,29]. Nevertheless, none of these conveniences exist in directed topology which is not unilaterally connected and such methods in [22,24,28,29] are inappropriate to be directly migrated to this situation. To deal with this problem, the authors supposed that each agent was equipped with a sensor to acquire the position information of any agent within its detection range [25,32,33], i.e., the interaction of position information is established by detection. However, the detection region was modeled as the circle in these works, which was ideal and overpriced. In actual applications, the performance and number of sensors that can be carried on a single unit are limited due to the factors such as cost and weight. Therefore, it is more general to consider that only partial individuals can carry detection equipment and treat the sensing zone as a sector, which is a challenge for designing an APF-based strategy.

Based on the above discussion, there are two challenges when applying APF-based strategy to formation tracking: (i) guarantee information interaction when the directed graph is not unilaterally connected and the sensing zone is modeled by sectors, (ii) redesign APF-based method which can eradicate the local minima or deadlock state so that it is not affected by the scale of the multi-agent system and the complexity of collision situation. Motivated by these demands, this paper aims at investigating time-varying formation tracking and collision avoidance problems for the multi-UGV system which consists of omnidirectional vehicles. The main contributions are summarized as follows.

- (1) By introducing detection behavior whose sensing zone is represented by the sector, the APF-based method is leveraged for collision avoidance during formation tracking when the directed graph is not unilaterally connected, which overcomes the dependence of APF-based method on the unilaterally connected graph when not considering the use of additional operations to obtain position information [22,30]. Different from the circular model utilized in [25,32,33], the area of detection is simplified into the sector in this work, which is more practical.
- (2) It is proved from the perspective of force balance that local minima or deadlock dilemmas can be eradicated when proposed composite potential field forces are adopted, which cannot be ensured in [22–27]. Compared with the proposals in [32,33], the one presented in this work is not easily affected by the complexity of the collision situation. In addition, to reduce the extra cost of detection, except for single-to-single detection, single-to-multiple detection is proposed in our research.
- (3) Time-varying formation tracking control protocol which is combined with composite potential field forces is designed for multi-UGV systems. Compared with [34–37], control protocol with the safety consideration is more suitable for practical application. In this research, the control protocol for a multi-UGV system is divided into two parts and designed separately. Benefit from the design of composite potential field forces and the eradication of local minima, potential field factors are not included in the construction of the Lyapunov function as [22,24], which makes the proof of convergence pretty neat.

The remainder of this paper is arranged as follows. Section 2 introduces the basic concepts and definitions. In Section 3, the composite potential field forces are proposed and the eradication of local minima or deadlock states is proved. The design of the control protocol and the corresponding analysis are provided in Section 4. Section 5 uses the simulation experiments to reveal the effectiveness of the proposed method. Section 6 gives the conclusion of this paper.

2. Preliminaries and problem statement

The dynamics property of omnidirectional vehicles is discussed in this section and the multi-UGV system is established based on basic graph theory.

2.1. Modeling an omnidirectional vehicle

An omnidirectional vehicle is depicted in Fig. 1. Suppose that R is the radius of each wheel, $l_1 \neq l_2$, $V_w = [V_{w1}, V_{w2}, V_{w3}, V_{w4}]^T$ is the velocity vector of four wheels, $V_b = [V_x, V_y, V_\theta]^T$ is the velocity vector in vehicle frame of reference. Each wheel is covered with rollers and

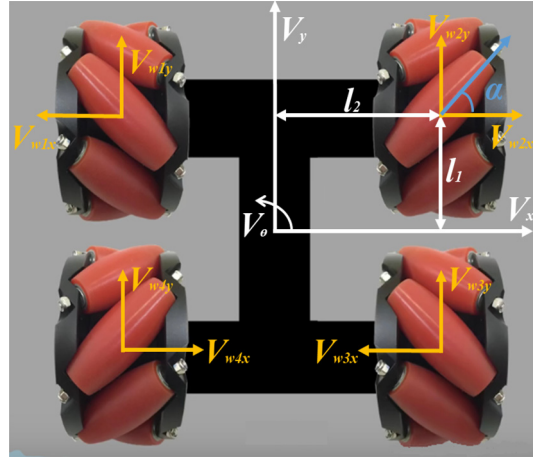


Fig. 1. Schematic of the considered vehicle model.

$\alpha = 45^\circ$ is denoted as the angle between the roller and V_x . $X = [x^w, y^w, \theta^w]^T$ is the coordinate and orientation of the vehicle in the world frame, $\theta^w \in [0, 2\pi)$. Inspired by [38], according to the forward and inverse kinematics model, one has

$$\begin{bmatrix} \dot{\theta}_1 \\ \dot{\theta}_2 \\ \dot{\theta}_3 \\ \dot{\theta}_4 \end{bmatrix} = RJ(\theta^w) \begin{bmatrix} \dot{x}^w \\ \dot{y}^w \\ \dot{\theta}^w \end{bmatrix}, \quad (1)$$

where $\theta_i, i = 1, 2, \dots, 4$ are rotation angles of four wheels and

$$J(\theta^w) = \begin{bmatrix} -\sqrt{2} \sin(\theta^w + \frac{\pi}{4}) & \sqrt{2} \cos(\theta^w + \frac{\pi}{4}) & l_1 + l_2 \\ \sqrt{2} \cos(\theta^w + \frac{\pi}{4}) & \sqrt{2} \sin(\theta^w + \frac{\pi}{4}) & -l_1 - l_2 \\ -\sqrt{2} \cos(\theta^w + \frac{\pi}{4}) & \sqrt{2} \sin(\theta^w + \frac{\pi}{4}) & -l_1 - l_2 \\ \sqrt{2} \sin(\theta^w + \frac{\pi}{4}) & \sqrt{2} \cos(\theta^w + \frac{\pi}{4}) & l_1 + l_2 \end{bmatrix}. \quad (2)$$

Suppose that the vehicle takes action in the horizontal plane, then the potential energy in Lagrangian function L can be ignored and the Euler-Lagrange equation is defined as

$$\frac{\partial}{\partial t} \left(\frac{\partial L}{\partial \dot{\theta}} \right) - \frac{\partial L}{\partial \theta} = \lambda - \frac{\partial C}{\partial \theta} - F(\dot{\theta}), \quad (3)$$

where $F(\dot{\theta}) = [\mu f_{w1} \text{sign}(\dot{\theta}_1), \mu f_{w2} \text{sign}(\dot{\theta}_2), \dots, \mu f_{w4} \text{sign}(\dot{\theta}_4)]^T$, $\text{sign}(\cdot)$ is the standard sign function, μ is the coefficient of static friction. $f_{wi}, i = 1, 2, \dots, 4$ represent the whole vehicle gravity borne by each wheel. $\lambda = [\lambda_1, \lambda_2, \dots, \lambda_4]^T$, $\theta = [\theta_1, \theta_2, \dots, \theta_4]^T$, $C = 0.5C_\mu(\dot{\theta}_1^2 + \dot{\theta}_2^2 + \dots + \dot{\theta}_4^2)$ is the energy cost due to static friction and C_μ is the viscous friction coefficient. From (3), one can obtain

$$\lambda = M\ddot{\theta} + C_\mu\dot{\theta} + F(\dot{\theta}), \quad (4)$$

$$\text{where } M = \begin{bmatrix} J_1 + J_2 + J_w & -J_2 & J_2 & J_1 - J_2 \\ -J_2 & J_1 + J_2 + J_w & J_1 - J_2 & J_2 \\ J_2 & J_1 - J_2 & J_1 + J_2 + J_w & -J_2 \\ J_1 - J_2 & J_2 & -J_2 & J_1 + J_2 + J_w \end{bmatrix}, \quad J_1 = \frac{mR^2}{8}, \quad J_2 =$$

$\frac{J_b R^2}{16(l_1 + l_2)}$, m is the mass of a vehicle, J_b is the moment of inertia about vehicle, J_w is the moment of inertia about each wheel.

Let $V = [\dot{x}^w, \dot{y}^w, \dot{\theta}^w]^T$, from (1) and (4), one can obtain

$$\dot{V} = RJ^+(\theta^w)[M^{-1}(\lambda - \frac{1}{R}C_\mu J(\theta^w)V - F(\dot{\theta}))] + \dot{J}^+(\theta^w)J(\theta^w)V, \quad (5)$$

where $J^+(\theta^w)J(\theta^w) = \mathbf{I}_3$, $\mathbf{I}_3$ is an identity matrix with dimension 3. The term $RJ^+(\theta^w)M^{-1}\lambda$ is chosen as the control input, then the omnidirectional vehicle can be modeled as

$$\begin{cases} \dot{X}(t) = V(t), \\ \dot{V}(t) = RJ^+(\theta^w)[M^{-1}(-\frac{1}{R}C_\mu J(\theta^w)V - F(\dot{\theta}))] + \dot{J}^+(\theta^w)J(\theta^w)V + u(t) \\ \quad = -f_o(t) + u(t), \end{cases} \quad (6)$$

where $u(t) = [u_x(t), u_y(t), u_\theta(t)]^T$, $f_o(t) = [f_x(t), f_y(t), f_\theta(t)]^T$ represents the nonlinear dynamic. Let $[u_x(t), u_y(t)] = u^a(t)$, $u_\theta(t) = u^b(t)$, $[f_x(t), f_y(t)] = f^a(t)$, $f_\theta(t) = f^b(t)$.

2.2. Graph theory

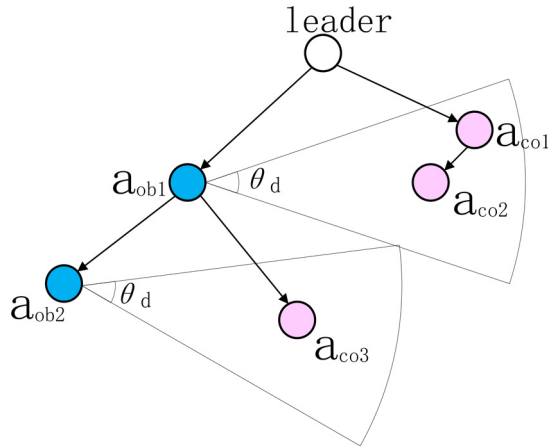
The communication network of a multi-UGV system containing N vehicles can be modeled by the directed graph $\mathcal{G} = \{\mathcal{V}, \mathcal{E}, \mathcal{W}\}$, where $\mathcal{V} = \{\mathcal{V}_1, \mathcal{V}_2, \dots, \mathcal{V}_N\}$ is the node set, $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ stands for the edge set, and the edge $(\mathcal{V}_i, \mathcal{V}_j)$ denotes that there is a directed edge from $\mathcal{V}_i$ to $\mathcal{V}_j$. $\mathcal{W} = [\omega_{ij}] \in \mathbb{R}^{N \times N}$ represents the adjacency matrix, $\omega_{ij} > 0$ only if $(\mathcal{V}_i, \mathcal{V}_j) \in \mathcal{E}$ while $\omega_{ij} = 0$ otherwise. Denote $\mathcal{D} \in \mathbb{R}^{N \times N}$ as the in-degree matrix of $\mathcal{G}$ and the graph Laplacian matrix $\mathcal{L}$ is defined as $\mathcal{L} = \mathcal{D} - \mathcal{W}$. Suppose that there exists a root node having directed paths to every other node, i.e., $\mathcal{G}$ has a spanning tree. A digraph $\mathcal{G}$ is unilaterally connected if, for every pair $\mathcal{V}_i, \mathcal{V}_j$ of vertices of $\mathcal{G}$, either $\mathcal{V}_i$ is reachable from $\mathcal{V}_j$ or $\mathcal{V}_j$ is reachable from $\mathcal{V}_i$ (or both).

2.3. Problem statement

Consider a multi-UGV system with 1 leader and $N - 1$ followers, where the leader follows the reference trajectory generated by a virtual leader whose orientation information does not affect formation tracking, and other vehicles follow the leader and move in formation while avoiding collisions among them. The virtual leader is a flexible definition, which can describe not only the desired reference trajectory but also an actual agent. If it represents a noncooperative target, its control input is usually unknown for all individuals in the multi-UGV system, and only the upper bound of this input is utilized to design the control protocol for each vehicle. Thus, it is more practical to consider the control input of the virtual leader as a bounded unknown signal. The virtual leader is modeled by

$$\dot{x}_l(t) = \tilde{A}x_l(t) + \tilde{B}u_l(t), \quad (7)$$

where $\tilde{A} = I_m \otimes \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$, $\tilde{B} = I_m \otimes \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, $x_l(t) = [x_l^w(t), \dot{x}_l^w(t), y_l^w(t), \dot{y}_l^w(t)]^T$, $u_l(t)$ is unknown control input of the virtual leader and $\|u_l(t)\|_2 \leq \tau$. I_m is an identity matrix with

Fig. 2. A network consists of a_{co} and a_{ob} .

dimension m which indicates the number of control inputs, $m = 2$ for omnidirectional vehicle. τ is a positive constant that represents the upper bound of $u_i(t)$. Based on (6), vehicles are described by

$$\dot{x}_i(t) = \tilde{A}x_i(t) + \tilde{B}(u_i(t) - f_i(t)), i \in 1, 2, \dots, N, \quad (8)$$

where $x_i(t) = [x_i^w(t), \dot{x}_i^w(t), y_i^w(t), \dot{y}_i^w(t)]^T$, $u_i(t) = (u_i^a(t))^T$ represents the control input of vehicle i , $f_i(t) = (f_i^a(t))^T$. The notation $i \in 1, 2, \dots, N$ is the serial number of each vehicle, e.g., $f^a(t)$ of vehicle i is represented by $f_i^a(t)$.

Consider the case where a directed graph with at least one spanning tree is utilized to describe the communication network, there may not be a directed path between some nodes. Thus, information interaction may be absent between some node pairs and collision avoidance cannot be fulfilled when not considering the use of additional operations to obtain position information. Notice that the leader may have to invest limited detection resources in observing the virtual leader, it is reasonable to assume the leader does not need to observe the followers. To ensure physical safety, divide N vehicles except for the leader into two categories. Let individuals equipped with extra sensors form a vehicle set a_{ob} , others form a vehicle set a_{co} . In a_{co} , vehicles can only obtain relative information through a communication topology network. Vehicles with detection behaviors in a_{ob} are observers, they can receive state information through both the communication topology and observation, which prevents the failure of collision avoidance caused by the lack of information interaction in unilaterally connected directed graphs. The detective area F is simplified as a sector whose center angle θ_d and radius d represent detection field angle and detection distance respectively. Normally, the orientation of the sensor corresponds to the orientation of the front of the vehicle and the angular bisector of the center angle coincides with V_y shown in Fig. 1. When vehicles continuously stay in the detective area of observer i , the observer can continuously receive the relative position and orientation of these vehicles by detection behavior. An observer can transfer information about its detected vehicles to its lower nodes in the directed graph. Besides, the influence on detection is not considered when other vehicles are between the observer and observed vehicles. A network with $a_{co} = \{a_{co1}, a_{co2}, a_{co3}\}$ and $a_{ob} = \{a_{ob1}, a_{ob2}\}$ is depicted in Fig. 2.

Based on the directed graph, the following Definition presents sufficient conditions for the multi-UGV system with detection behavior and collision avoidance to safely accomplish time-varying formation tracking.

Definition 1. For any given bounded initial state, if

$$\lim_{t \rightarrow \infty} (x_i(t) - h_i(t) - x_l(t)) = 0, i = 1, 2, \dots, N, \quad (9)$$

$$S\{a_{od}^q\} \subseteq S\{F_q\}, q \in a_{ob}, \quad (10)$$

$$D_{ze}^b > rr, z \neq e, z, e = 1, 2, \dots, N, \quad (11)$$

where $h_i(t) = [x_i^h(t), \dot{x}_i^h(t), y_i^h(t), \dot{y}_i^h(t)]^T$ is the position and velocity command for vehicle i relative to virtual leader. a_{od}^q is the vehicle set whose individuals are detected by observer q , $S\{a_{od}^q\}$ represents the space occupied by a_{od}^q . F_q represents the detective area of observer q , $S\{F_q\}$ is the space occupied by F_q . D_{ze}^b is the distance between vehicle z and e , rr is the minimum distance allowed between any two vehicles. Then the multi-UGV system with directed graph, detection behavior, and collision avoidance is said to safely accomplish time-varying formation tracking specified by $h(t) = [h_1^T(t), h_2^T(t), \dots, h_N^T(t)]^T$.

Remark 1. Sufficient conditions for achieving formation tracking, continuous observation, and collision avoidance for multi-UGV systems on a directed graph are given by Definition 1. Specifically, (9) is fulfilled by position and velocity control, (10) and (11) are ensured by yaw control and APF-based method, which will be discussed later.

3. Composite potential field forces for multi-UGV system with collision avoidance and detection behavior

To meet the requirements of collision avoidance and detection behavior, different potential field forces are designed in this section. As an improved APF-based method, cross-repulsive potential field force is designed to prevent collisions while eliminating local minima or dead-lock dilemmas. Then different composite potential field forces are presented in two different situations: (i) single-to-single detection where a vehicle can only observe one vehicle, (ii) single-to-multiple detection where a vehicle has the ability to observe multiple individuals at the same time.

3.1. Cross-repulsive potential field force

For multi-agent systems, the local minima or deadlock dilemmas of APF-based strategy have been bothering researchers, which results in the low practicability of the traditional APF-based method. Denote $\vec{f}_c(f_{cx}, f_{cy}, f_{cz})$ as the force which is created by formation tracking control protocol, $\mathbf{F}$ is the force which is generated by APF. The problem above usually occurs when the sum of all control elements equals zero and remains unchanged, i.e., the resultant force consisting of $\vec{f}_c$ and $\mathbf{F}$ stabilizes at zero before the whole system converges to

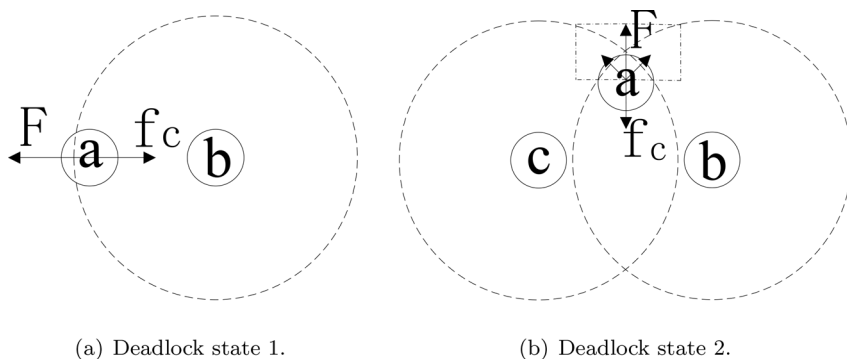


Fig. 3. Mechanism of deadlock.

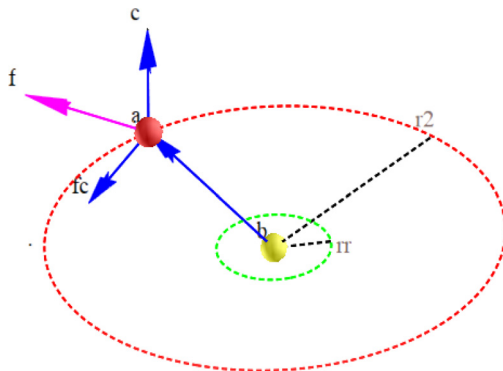


Fig. 4. Improved potential field force.

desired state. As shown in Fig. 3, when agent a enters the region of traditional APF which is indicated by a dotted circle and around obstacles, F would be generated in the opposite direction of f_c . As the distance between a and obstacles decreases, the resultant force may weaken to zero and even push a away, which results in the oscillation of a or deadlock of the system.

To eradicate local minima and deadlock states, an APF-based strategy is designed based on cross-product. Suppose that the radius of an individual is r_1 and agent $a(x_1, y_1, z_1)$ needs to avoid agent $b(x_0, y_0, z_0)$. $rr = 4r_1$ is the minimum distance allowed between a and b . Let the repulsive potential field circle around agent b and define $r_2 \geq 3rr$ as the radius of the repulsive region. ϕ denotes the plane which contains b , a and $\vec{f}_c$. Besides, it is stipulated that only one individual in each pair of agents can be affected by the potential field. Take a as the analysis object, when $\vec{ba}$ is not collinear with $\vec{f}_c$ and $\|\vec{f}_c\|_2 \neq 0$, the improved potential field force is depicted in Fig. 4. Denote $\vec{f} = [f_1 \ f_2 \ f_3]^T$ as the unit vector of cross-repulsive

potential field force, it can be obtained that

$$\begin{aligned}\vec{f} &= \frac{\vec{f}_c \times (\vec{b}\vec{a} \times \vec{f}_c)}{\|\vec{f}_c \times (\vec{b}\vec{a} \times \vec{f}_c)\|_2} \\ &= \frac{\begin{bmatrix} ((x_1 - x_0)f_{cy} - (y_1 - y_0)f_{cx})f_{cy} - ((z_1 - z_0)f_{cx} - (x_1 - x_0)f_{cz})f_{cz} \\ ((y_1 - y_0)f_{cz} - (z_1 - z_0)f_{cy})f_{cz} - ((x_1 - x_0)f_{cy} - (y_1 - y_0)f_{cx})f_{cx} \\ ((z_1 - z_0)f_{cx} - (x_1 - x_0)f_{cz})f_{cx} - ((y_1 - y_0)f_{cz} - (z_1 - z_0)f_{cy})f_{cy} \end{bmatrix}}{\left\| \begin{bmatrix} ((x_1 - x_0)f_{cy} - (y_1 - y_0)f_{cx})f_{cy} - ((z_1 - z_0)f_{cx} - (x_1 - x_0)f_{cz})f_{cz} \\ ((y_1 - y_0)f_{cz} - (z_1 - z_0)f_{cy})f_{cz} - ((x_1 - x_0)f_{cy} - (y_1 - y_0)f_{cx})f_{cx} \\ ((z_1 - z_0)f_{cx} - (x_1 - x_0)f_{cz})f_{cx} - ((y_1 - y_0)f_{cz} - (z_1 - z_0)f_{cy})f_{cy} \end{bmatrix} \right\|_2}.\end{aligned}\quad (12)$$

Denote θ_{fo} as the angle between $\vec{b}\vec{a}$ and $\vec{f}$, then

$$\begin{aligned}\cos(\theta_{fo}) &= \frac{\vec{f}\vec{b}\vec{a}}{\|\vec{f}\|_2\|\vec{b}\vec{a}\|_2} \\ &= \frac{\vec{f}_c \times (\vec{b}\vec{a} \times \vec{f}_c)\vec{b}\vec{a}}{\|\vec{f}_c \times (\vec{b}\vec{a} \times \vec{f}_c)\|_2\|\vec{b}\vec{a}\|_2} \\ &= \frac{(x_1 - x_0)^2 f_{cy}^2 + (y_1 - y_0)^2 f_{cx}^2 - 2(x_1 - x_0)(y_1 - y_0)f_{cy}f_{cx}}{\|\vec{f}_c \times (\vec{b}\vec{a} \times \vec{f}_c)\|_2\|\vec{b}\vec{a}\|_2} \\ &\quad + \frac{(x_1 - x_0)^2 f_{cz}^2 + ((z_1 - z_0))^2 f_{cx}^2 - 2(x_1 - x_0)(z_1 - z_0)f_{cz}f_{cx}}{\|\vec{f}_c \times (\vec{b}\vec{a} \times \vec{f}_c)\|_2\|\vec{b}\vec{a}\|_2} \\ &\quad + \frac{(z_1 - z_0)^2 f_{cy}^2 + (y_1 - y_0)^2 f_{cz}^2 - 2(y_1 - y_0)(z_1 - z_0)f_{cy}f_{cz}}{\|\vec{f}_c \times (\vec{b}\vec{a} \times \vec{f}_c)\|_2\|\vec{b}\vec{a}\|_2} \\ &\geq 0.\end{aligned}\quad (13)$$

For the situation where $\vec{b}\vec{a}$ is collinear with $\vec{f}_c$ and $\|\vec{f}_c\|_2 \neq 0$, if $\vec{b}\vec{a}$ goes in the same direction as $\vec{f}_c$, an downward unit vector which is perpendicular to the plane ϕ can be constructed. When $\vec{b}\vec{a}$ and $\vec{f}_c$ are in the opposite direction, an upward unit vector that is perpendicular to the plane ϕ can be constructed. Denote $\vec{c} = (c_x, c_y, c_z)$ as the vector which is perpendicular to ϕ , the corresponding improved potential field force is shown in Fig. 5 and one has

$$\vec{f} = \frac{\vec{f}_c \times \vec{c}}{\|\vec{f}_c \times \vec{c}\|_2}.\quad (14)$$

As for $\|\vec{f}_c\|_2 = 0$, the establishments of unit vectors when a approaches to b or moves away from b are similar to (14), which are described by

$$\vec{f} = \frac{\vec{a}\vec{b} \times \vec{c}}{\|\vec{a}\vec{b} \times \vec{c}\|_2}\quad (15)$$

and

$$\vec{f} = \frac{\vec{b}\vec{a} \times \vec{c}}{\|\vec{b}\vec{a} \times \vec{c}\|_2},\quad (16)$$

where $\vec{c}$ is upward in (15) and downward in (16).

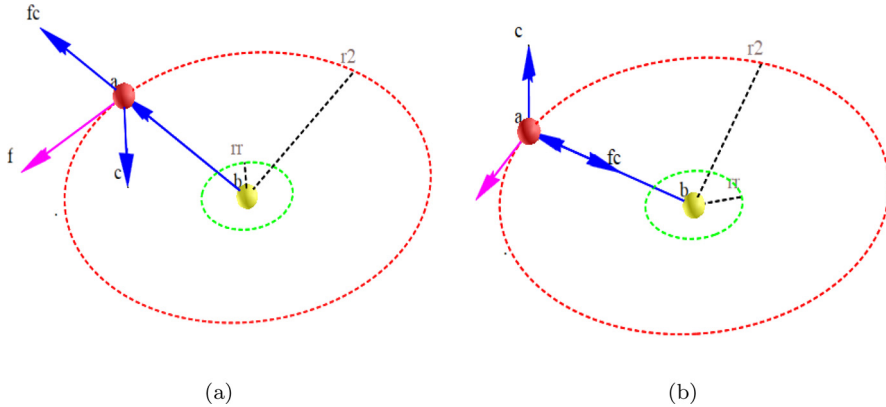


Fig. 5. Special cases of the improved potential field force.

From (12)–(16), one has $\theta_{fo} \in [0, \frac{\pi}{2}]$, namely, the improved potential field force can move a away from b , which prevents the collision between them.

For UGVs, $f_3 = z_0 = z_1 = f_{cz} = 0$, then the repulsive potential field force is designed as

$$\mathbf{F} = \begin{cases} \frac{k_1(r_2^2 - rr^2)(r_2^2 - \|\vec{ba}\|_2^2)}{(\|\vec{ba}\|_2^2 - rr^2)^3} (f_1\mathbf{i} + f_2\mathbf{j}), & rr < \|\vec{ba}\|_2 \leq r_2, \\ 0, & \|\vec{ba}\|_2 > r_2, \end{cases} \quad (17)$$

where k_1 is the weight of repulsive force, $\mathbf{i}$ and $\mathbf{j}$ represent imaginary unit. Note that $\mathbf{F}$ is continuous at $\|\vec{ba}\|_2 = r_2$ and $\mathbf{F}$ monotonically decreases with the increase of $\|\vec{ba}\|_2$ when $\|\vec{ba}\|_2 > rr$. Besides, $\|\mathbf{F}\|_2 \rightarrow \infty$ when $\|\vec{ba}\|_2 \rightarrow rr+$, which means the effect of the potential field would be much greater than that of formation control protocol, i.e., collision avoidance is ensured.

Remark 2. The cross-repulsive potential field force could get rid of the dilemma encountered by traditional APFs. An example is shown in Fig. 6, where b and c are two vehicles that should be avoided by a . b cannot take the initiative to avoid collision and a moves along the passage in the middle of two vehicles. When b is far from c , the cross-repulsive forces F_1 and F_2 created by b and c are perpendicular to $\vec{f}_c$, then repulsive force equilibrium is reached. Thus, a could pass through the area without being influenced. When b is close to c , the passage between two vehicles is blocked. According to $r_2 > 3rr$, c will gradually move away from b under the influence of repulsive force F_3 which is created by b , and the passage will be unblocked for a to pass. Besides, when the position of c changes, a will not stay on the centerline of b and c , then repulsive force equilibrium is broken, and the failure of collision avoidance caused by the continuous cancellation of potential field forces will not occur.

Remark 3. Classic resultant potential field force only depends on the relative position of agents [22], which may result in a deadlock state. However, the cross-repulsive potential field force is generated by relative position and formation control protocol, which can effectively eliminate the local minima.

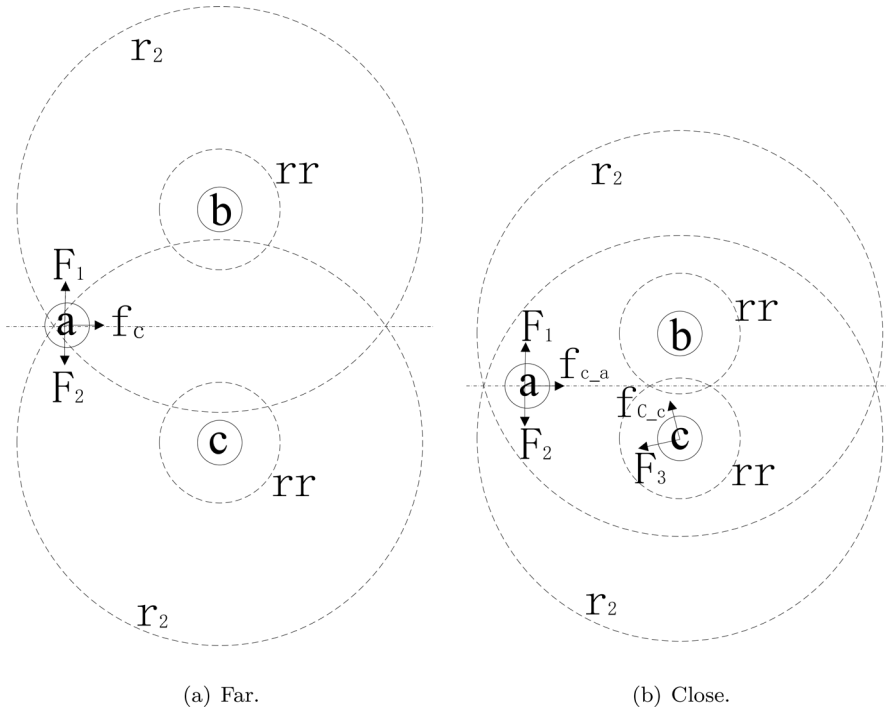


Fig. 6. Analysis of cross-repulsive potential field force.

3.2. Composite potential field force in single-to-single detection

In the situation where a vehicle $a(x_0, y_0, z_0)$ can only detect one vehicle $b(x_1, y_1, z_1)$, to make b stay in the range of sector, two constraints are required: (i) the angular bisector of θ_d points to b all the time, (ii) the distance between a and b is less than d but greater than rr . Note that the yaw control and position control of the omnidirectional vehicle are decoupled, for ease of analysis, constraint (i) is achieved by the control protocol, constraint (ii) is fulfilled by composite potential field force as shown in Fig. 7. In the first region which is composed of a circle with radius r_2 , the vehicle is repelled by the repulsive potential field force from b . In the third region which is denoted by a circle with radius $r_3 \geq 3r_2$, attractive potential field force f_d takes effect to constrain the maximum distance between a and b .

Denote k_2 as the weight of attractive force, then attractive potential field force from a to b is given by

$$\mathbf{f}_d = \begin{cases} k_2 \frac{\|\vec{ba}\|_2^2 - r_3^2}{(d^2 - \|\vec{ba}\|_2^2)^2 \|\vec{ba}\|_2} \left((x_1 - x_0)\mathbf{i} + (y_1 - y_0)\mathbf{j} \right), & r_3 \leq \|\vec{ba}\|_2 < d, \\ 0, & 0 < \|\vec{ba}\|_2 < r_3, \end{cases} \quad (18)$$

where $\mathbf{f}_d$ alters with respect to $\|\vec{ba}\|_2$ when $r_3 \leq \|\vec{ba}\|_2 < d$. Note that $\|\mathbf{f}_d\|_2 \rightarrow \infty$ when $\|\vec{ba}\|_2 \rightarrow d^-$, which means the attractive force would be extremely large to fulfill the constraint of detection distance. In the area where $r_2 < \|\vec{ba}\|_2 < r_3$, no potential field force is generated. Besides, the cross-repulsive potential field force takes over when $\|\vec{ba}\|_2 \leq r_2$.

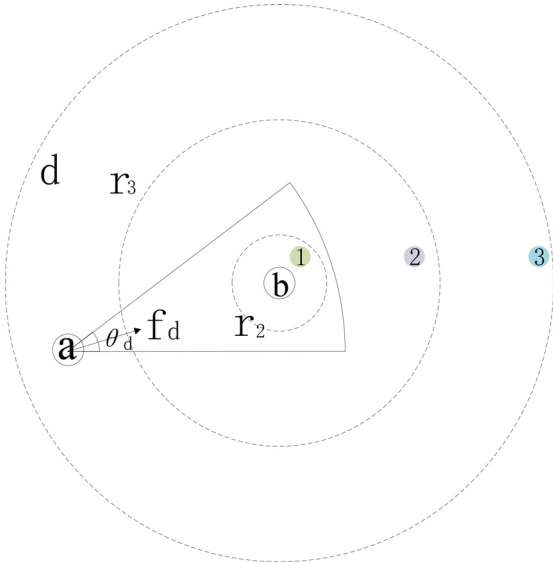


Fig. 7. Situation of single-to-single detection.

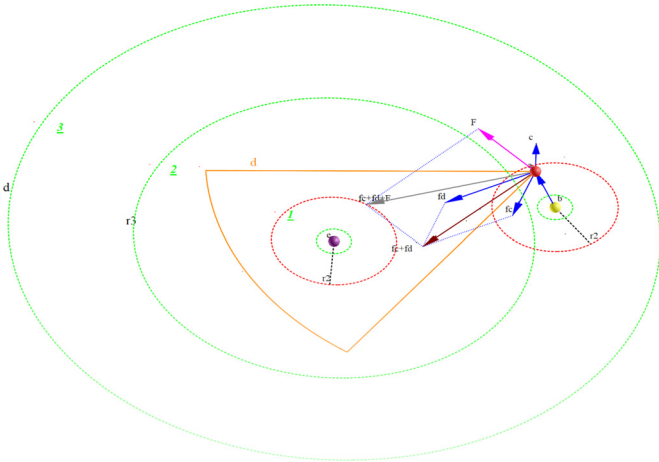


Fig. 8. Analysis of attractive and repulsive forces.

Remark 4. Suppose that vehicle a is subject to attractive force from vehicle e and repulsive force from vehicle b , a deadlock state may occur when the direction of attractive force is not strictly perpendicular to the control protocol. To solve the above problem, $\vec{f}_c$ in Section 3.1 is replaced by $\vec{f}_c + \vec{f}_d$ and the analysis of composite potential field force is shown in Fig. 8, where the resultant force is formed by control force $\vec{f}_c$, attractive force $\vec{f}_d$ and repulsive force $\vec{F}$, $\vec{F}$ is generated by $\vec{f}_c + \vec{f}_d$ and $\vec{ba}$. Let $\vec{f}$ be the unit vector of $\vec{F}$ in single-to-single

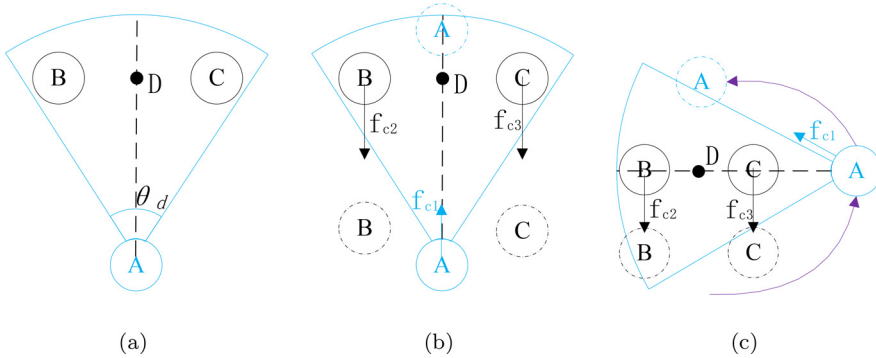


Fig. 9. Analysis of single-to-multiple situation.

detection, when $\vec{f}_c + \vec{f}_d \neq \vec{0}$ and $\vec{f}_c + \vec{f}_d$ is not collinear with $\vec{ba}$, $\vec{f}$ is given by

$$\begin{bmatrix} f_1 \\ f_2 \\ f_3 \end{bmatrix} = \frac{(\vec{f}_c + \vec{f}_d) \times \vec{c}}{\|(\vec{f}_c + \vec{f}_d) \times \vec{c}\|_2}, \quad (19)$$

where $\vec{c} = \vec{ba} \times (\vec{f}_c + \vec{f}_d)$. $\vec{F}$ can be obtained by substituting (19) into (17) and the resultant force is $\vec{f}_c + \vec{f}_d + \vec{F}$.

When $\vec{f}_c + \vec{f}_d$ is collinear with $\vec{ba}$, the construction of $\vec{f}$ is similar to (14). If $\vec{f}_c + \vec{f}_d = \vec{0}$, please refer to (15) and (16) to calculate $\vec{f}$.

3.3. Composite potential field force in single-to-multiple detection

For single-to-multiple detection, multiple vehicles can be detected by one vehicle. Inspired by single-to-single detection, one could utilize attractive force to limit the distances between vehicles and their observers. However, it is impossible to observe multiple vehicles continuously only by the yaw control protocol exerted on observers. In Fig. 9(a), D is the center of vehicle B and vehicle C . To make vehicle A observe vehicle B and vehicle C at the same time, the angular bisector of θ_d points to D . In Fig. 9(b), the final states of three vehicles are marked by dotted circles. Under the influence of formation control protocol f_{c1} , f_{c2} and f_{c3} , vehicle A moves between B and C , which results in the failure of continuous detection. An ideal moving process is shown in Fig. 9(c), where vehicle A bypasses two vehicles and continuous detection is ensured.

To accomplish the detour process shown in Fig. 9(c), composite potential field force in single-to-multiple detection is composed of four parts, including the gathering force, the attractive force of detection, the repulsive force of detection, and individual repulsive force. Details are elaborated as follows.

The concept of agent cluster is introduced before the discussion of gathering force. An agent cluster is a vehicle set whose individuals are observed by the same observer. Denote vehicle $A(x_A, y_A, z_A)$ as an observer, then the agent cluster detected by A can be nominated as $A_d = \{A_1, A_2, \dots, A_n\}$, where n is the sum of vehicles contained in A_d . Besides, the topology of an agent cluster has at least one directed spanning tree. Based on above knowledge, a vehicle $A_i(x_i, y_i, z_i)$, $A_i \in A_d$ which transfers information to other vehicles $A_j(x_j, y_j, z_j)$, $A_j \neq$

$A_i, A_j \in A_d$ can be obtained. Assume A_j has no detection behavior, the gathering force is designed as

$$\vec{f}_{M1} = \begin{cases} k_3 \frac{\|\vec{A_i A_j}\|_2^2 - r_3^2}{(dd_1^2 - \|\vec{A_i A_j}\|_2^2)^2 \|\vec{A_i A_j}\|_2} \left((x_i - x_j)\mathbf{i} + (y_i - y_j)\mathbf{j} \right), & r_3 \leq \|\vec{A_i A_j}\|_2 < dd_1, \\ 0, & 0 < \|\vec{A_i A_j}\|_2 < r_3, \end{cases} \quad (20)$$

where dd_1 represents the upper bound of the gathering force region, and k_3 is the weight of the gathering force. It follows from (20) that all vehicles in A_d will gather in a cluster with radius dd_1 .

To make the agent cluster stay in the detection range of its observer, the attractive force of detection is utilized to constrain the distance between A_i and A , which is given by

$$\vec{f}_{M3} = \begin{cases} k_4 \frac{\|\vec{A_i A}\|_2^2 - dd_2^2}{(dd_2^2 - \|\vec{A_i A}\|_2^2)^2 \|\vec{A_i A}\|_2} \left((x_i - x_A)\mathbf{i} + (y_i - y_A)\mathbf{j} \right), & dd_3 \leq \|\vec{A_i A}\|_2 < dd_2, \\ 0, & 0 < \|\vec{A_i A}\|_2 < dd_3, \end{cases} \quad (21)$$

where $dd_2 = d - dd_1$, dd_3 represents the lower bound of attractive force region, k_4 is the weight of attractive force.

If the observer moves too close to its agent cluster, partial vehicles inside the agent cluster may move out of the detection sector. Inspired by the design of cross-repulsive force, view the agent cluster as an individual with a larger radius dd_1 whose center is vehicle A_i , then the detour process can be established between the observer and agent cluster. Denote $\vec{f}_{cA}$ as the control force which is created by the formation control protocol of A . When $\vec{A_i A}$ is not collinear with $\vec{f}_{cA}$ and $\vec{f}_{cA} \neq \vec{0}$, the unit vector which represents the repulsive force of detection is constructed as

$$\begin{aligned} \vec{f}_m &= \frac{\vec{f}_{cA} \times (\vec{A_i A} \times \vec{f}_{cA})}{\|\vec{f}_{cA} \times (\vec{A_i A} \times \vec{f}_{cA})\|_2} \\ &= \begin{bmatrix} f_{a1} \\ f_{a2} \\ f_{a3} \end{bmatrix}. \end{aligned}$$

Then the repulsive force of detection is designed as

$$\vec{f}_{M2} = \begin{cases} k_5 \frac{(r_{rmax}^2 - r_{rmin}^2)(r_{rmax}^2 - \|\vec{A_i A}\|_2^2)}{(\|\vec{A_i A}\|_2^2 - r_{rmin}^2)^3 \|\vec{A_i A}\|_2} (f_{a1}\mathbf{i} + f_{a2}\mathbf{j}), & r_{rmin} < \|\vec{A_i A}\|_2 \leq r_{rmax}, \\ 0, & \|\vec{A_i A}\|_2 > r_{rmax}, \end{cases} \quad (22)$$

where $r_{rmin} = \frac{dd_1}{\sin(\theta_d/2)}$, r_{rmax} is the upper bound of repulsive force region, k_5 is the weight of repulsive force. When $\vec{A_i A}$ is collinear with $\vec{f}_{cA}$, the calculation of $\vec{f}_m$ is similar to (14). Besides, $\vec{f}_m$ can be obtained according to (15) and (16) if $\vec{f}_{cA} = \vec{0}$. When vehicle $A_k, A_k \notin A_d \wedge A_k \neq A$ enters the region of repulsive force, the same repulsive mechanism is effective between agent cluster and A_k . Distinctively, information interaction between A_i and A_k is based on communication topology. Finally, the design of individual repulsive force is as same as that of the previous cross-repulsive potential field force, which will not be discussed here.

Remark 5. In single-to-multiple detection, gathering force is used to form multiple vehicles detected by the same observer into a cluster, the attractive force of detection is utilized to

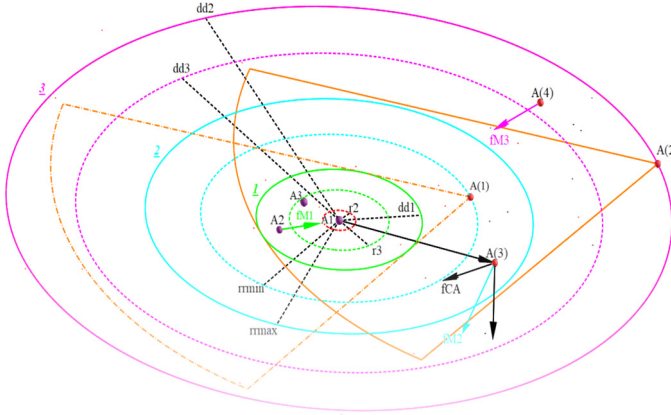


Fig. 10. Analysis of composite potential field force in single-to-multiple detection.

constrain the maximum distance while the repulsive force of detection is designed to limit the minimum distance between the agent cluster and its observer. Based on the composite potential field force and the yaw control protocol which will be introduced later, multiple vehicles observed by the same observer can be kept in its detection sector, and collision avoidance is fulfilled by individual repulsive force. When $A_d = \{A_1, A_2, A_3\}$, A_2 and A_3 can receive information from A_1 by communication topology, the analysis of composite potential field force is shown in Fig. 10, where A_2 and A_3 form an agent cluster around A_1 under the influence of gathering force $\vec{f}_{M1}$, which is marked by green colour. $A(1)$ and $A(2)$ indicate the closest and farthest positions allowed for observer. A vehicle at $A(3)$ will be pushed away from the agent cluster by repulsive detecting force $\vec{f}_{M2}$ and attractive detecting force will take effect to make the vehicle at $A(4)$ approach to agent cluster. Besides, physical safety is guaranteed by individual repulsive force.

Remark 6. $dd_1 - r_3$, $r_{max} - r_{min}$, $dd_3 - r_{max}$, and $dd_2 - dd_3$ are effective distances of different potential field forces which can reflect the change rates of forces. Specifically, change rates increase as the effective distances decrease. In actual application, the values of effective distances are subject to d , θ_d , and the maneuverability of vehicles. $k_1, \dots, k_5$ are parameters that represent the weights of different components in composite potential field forces. They can be viewed as the auxiliary parameters of corresponding effective distances to help regulate the change rates of forces. According to the construction principle of composite potential field force, these gains have no impact on the conclusion of eradicating the local minima or deadlock state. Therefore, the variations of these gains only affect the intensities of different potential field forces and it is only necessary to ensure that the effect of potential field forces does not exceed the upper limit that an individual's mobility can withstand after configuring these parameters.

3.4. Analysis of deadlock state based on composite potential field force

For a multi-UGV system integrated with the proposed APF-based method, the following results can be derived.

Theorem 1. Suppose that potential field forces are described by (12)–(22), then there are no local minima or deadlock states for the multi-UGV system with collision avoidance and detection behavior.

Proof. Consider the situation where vehicles are motivated by formation control protocol and repulsive potential field force. Denote $\mathbf{F} = [\vec{F}_1, \vec{F}_2, \dots, \vec{F}_N]$ and $\mathbf{f} = [\vec{f}_{c1}, \vec{f}_{c2}, \dots, \vec{f}_{cN}]$ as the potential field force and formation control force of multi-UGV system respectively which consists of N vehicles. If the local minima or deadlock state exists, then $\exists t > 0$, one has $\vec{0} \in \mathbf{F} + \mathbf{f}$ when $\vec{f}_{ci} \neq \vec{0} \vee \vec{F}_i \neq \vec{0}, i \in 1, 2, \dots, N$. However, according to (12) and (14)–(17), it can be obtained that $\vec{f}_c \perp \vec{f}$ and $\vec{f}_{ci} \perp \vec{F}_i$, then $\vec{0} \notin \mathbf{F} + \mathbf{f}$ when $\vec{f}_{ci} \neq \vec{0} \vee \vec{F}_i \neq \vec{0}, i \in 1, 2, \dots, N$, which is contradicted to the original assumption. Therefore, the original assumption is not tenable.

In single-to-single detection where vehicles are motivated by a formation control protocol, repulsive potential field force, and attractive potential field force, denote $\mathbf{f}_d = [\vec{f}_{d1}, \vec{f}_{d2}, \dots, \vec{f}_{dN}]$ as the attractive force of the multi-UGV system. If the local minima or deadlock state exists, then $\exists t > 0$, one has $\vec{0} \in \mathbf{F} + \mathbf{f} + \mathbf{f}_d$ when $\vec{f}_{ci} + \vec{f}_{di} \neq \vec{0} \vee \vec{F}_i \neq \vec{0}, i \in 1, 2, \dots, N$. From (19), $\vec{f} \perp (\vec{f}_c + \vec{f}_d)$ and $\vec{F}_i \perp (\vec{f}_{ci} + \vec{f}_{di})$ can be obtained, then $\vec{0} \notin \mathbf{F} + \mathbf{f} + \mathbf{f}_d$ when $\vec{f}_{ci} + \vec{f}_{di} \neq \vec{0} \vee \vec{F}_i \neq \vec{0}, i \in 1, 2, \dots, N$, which is inconsistent with the original assumption. Thus, the original assumption is not tenable.

For single-to-multiple detection, vehicles are motivated by the repulsive force of detection $\mathbf{F}_{M2} = [\vec{f}_{M2}^1, \vec{f}_{M2}^2, \dots, \vec{f}_{M2}^N]$, gathering force $\mathbf{F}_{M1} = [\vec{f}_{M1}^1, \vec{f}_{M1}^2, \dots, \vec{f}_{M1}^N]$, the attractive force of detection $\mathbf{F}_{M3} = [\vec{f}_{M3}^1, \vec{f}_{M3}^2, \dots, \vec{f}_{M3}^N]$, formation control protocol $\mathbf{f}$ and individual repulsive force $\mathbf{F}$. If local minima or deadlock state exists, for $i \in 1, 2, \dots, N, \exists t > 0$, when $\vec{f}_{ci} + \vec{f}_{M1}^i \neq \vec{0} \vee \vec{F}_i \neq \vec{0}, \vec{f}_{ci} \neq \vec{0} \vee \vec{F}_i + \vec{f}_{M2}^i \neq \vec{0}$ or $\vec{f}_{ci} + \vec{f}_{M3}^i \neq \vec{0} \vee \vec{F}_i \neq \vec{0}$ holds, one can obtain $\vec{0} \in \mathbf{f} + \mathbf{F}_{M1} + \mathbf{F} \vee \vec{0} \in \mathbf{f} + \mathbf{F}_{M2} + \mathbf{F} \vee \vec{0} \in \mathbf{f} + \mathbf{F}_{M3} + \mathbf{F}$. Similar to the previous discussion, based on (19)–(22), one has $\vec{F}_i \perp (\vec{f}_{M3}^i + \vec{f}_{ci})$, $\vec{F}_i \perp (\vec{f}_{M1}^i + \vec{f}_{ci})$ and $\vec{f}_{ci} \perp (\vec{F}_i + \vec{f}_{M2}^i), i \in 1, 2, \dots, N$. Namely, when $\vec{f}_{ci} + \vec{f}_{M1}^i \neq \vec{0} \vee \vec{F}_i \neq \vec{0}, \vec{f}_{ci} \neq \vec{0} \vee \vec{F}_i + \vec{f}_{M2}^i \neq \vec{0}$ or $\vec{f}_{ci} + \vec{f}_{M3}^i \neq \vec{0} \vee \vec{F}_i \neq \vec{0}$ holds, $\vec{0} \notin \mathbf{f} + \mathbf{F}_{M1} + \mathbf{F}, \vec{0} \notin \mathbf{f} + \mathbf{F}_{M2} + \mathbf{F}$ or $\vec{0} \notin \mathbf{f} + \mathbf{F}_{M3} + \mathbf{F}$ can be guaranteed, which is contradicted to the original assumption. This completes the proof of Theorem 1. $\square$

Remark 7. By analyzing the resultant force, Theorem 1 is proved based on the reduction to absurdity. Note that $\vec{f}_{ci} + \vec{f}_{di} \equiv \vec{0}$ holds only when formation configuration is unreasonable. Besides, $\vec{F}_i + \vec{f}_{M2}^i \equiv \vec{0}$ does not hold due to the repulsive force of detection is effective between agent cluster and other vehicles, the analysis is similar to it in Fig. 6. Compared with [22–27], local minima or deadlock dilemma is theoretically avoided. Denote D_{ze}^b as the distance between vehicle z and vehicle e where $z, e \in 1, 2, \dots, N, z \neq e$. Based on Theorem 1, for each agent cluster A_d whose central vehicle is A_i and observed by vehicle A , one has: $rr < \|\vec{A_i A_j}\|_2 < dd_1, A_j \in A_d, A_i \neq A_j; r_{min} < \|\vec{A_i A}\|_2 < dd_2, A \in a_{ob}; r_{min} < \|\vec{A_i A_k}\|_2, A_k \notin A_d \wedge A_k \neq A; D_{ze}^b > rr$.

Remark 8. For the collision avoidance strategy proposed in [29], a temporary state is selected to generate an extra attractive force that helps an agent circumnavigate another. However, it is time-consuming to judge whether the problem of local minima occurs, and the selection of the temporary state is susceptible to the system scale. For the repulsive vector field-based method presented in [32,33], the undesired saddle point (local minimum) is removed by carefully

redesigning the potential field function. However, it is complicated to design this function when an agent is in multiple repulsive potential fields at the same time and several undesired saddle points exist in phase portrait. In contrast, the local minima dilemma is eradicated from the perspective of reconstructing potential field force in this work and no extra operations are required when the scale of the multi-agent system is large. Namely, the improved strategy is not affected by the complexity of the collision situation. In addition, this work focuses on a more general situation where the directed graph is not unilaterally connected and the detection region is considered a sector. In this case, none of the methods in [29,32,33] can be directly leveraged for collision avoidance. Furthermore, the effect of formation tracking would not be weakened because the repulsive force is designed to be perpendicular to the formation tracking control force, which could be regarded as an advantage.

4. Formation tracking control for multi-UGV system

To achieve detection behavior at a low cost, the generation algorithm of a complete Laplacian matrix which represents communication and detection behavior is proposed in this section. Then protocols are designed from the perspectives of position and velocity control, and yaw control respectively.

Influenced by detection behavior, new information links are established between the observers and the vehicles which are observed. For vehicles with detection behaviors, the corresponding diagonal elements in matrix $\mathcal{D}$ are changed, and the adjacency matrix $\mathcal{W}$ is modified too. Let $\mathcal{W}'$ denote new adjacency matrix, $\mathcal{D}'$ is new in-degree matrix, the complete Laplacian matrix is $\mathcal{L}' = \mathcal{D}' - \mathcal{W}'$.

4.1. Generation of complete laplacian matrix

The single-to-single detection can be viewed as a special case of single-to-multiple detection, then the following discussion is based on single-to-multiple detection. To lower the cost of detection, an algorithm that could minimize the number of vehicles with detection behaviors is proposed. For a multi-UGV system with N vehicles, the number of vehicles contained in one agent cluster is up to n . Denote $C_d = \{(a_i, a_j) | i \neq j, i, j \in 1, 2, \dots, N\}$ as the set of vehicle pairs where each vehicle pair is connected by communication topology directly or indirectly, vehicle pair (a_i, a_j) is equivalent to vehicle pair (a_j, a_i) . Let $C_{na} = \{(a_i, a_j) | i \neq j, i, j \in 1, 2, \dots, N\}$ denote the set of all vehicle pairs. Then $C_n = C_{na} - C_d$ represents the information links that need to be established by detection behaviors. Let C_{new} represent the set of new information links which will be constructed when an agent cluster and its observer are selected. The vehicle set whose individuals could be selected as observers is denoted as D_c . Let $DM = [dm_{ij}] \in \mathbb{R}^{N \times N}$ be the detection matrix, $dm_{ij} = 1$ if and only if vehicle i is the observer of the agent cluster whose center is vehicle j while $dm_{ij} = 0$ otherwise.

Algorithm 1. Based on the original communication topology, the complete Laplacian matrix can be generated by the following five steps.

Step 1: Input the original communication topology.

Step 2: Generate C_d and C_n according to $\mathcal{L}$.

Step 3: Select an agent cluster and its observer, update C_{new} . The selection process is based on the following principles: (i) an agent cluster should contain as many vehicles as possible but less than n , (ii) the communication topology of an agent cluster has a directed spanning tree, (iii) elements in C_{new} should be as much as possible.

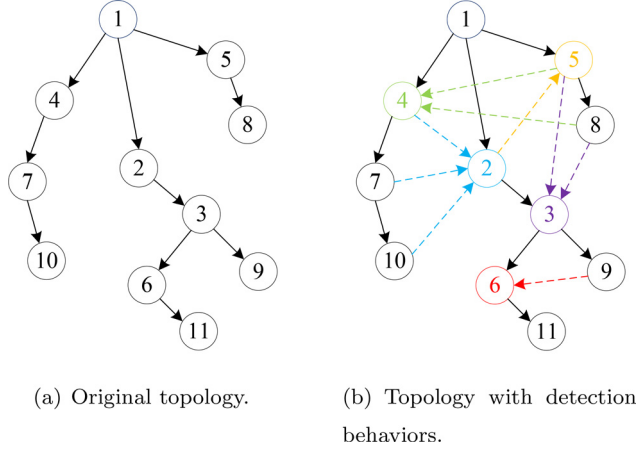


Fig. 11. Generation of detection behavior.

Step 4: Update $\mathcal{D}'$, $\mathcal{W}'$ and D_c , $C_n = C_n - C_{new}$. If $C_n \neq \emptyset \wedge D_c \neq \emptyset$, back to Step3; if $C_n \neq \emptyset \wedge D_c = \emptyset$, complete Laplacian matrix cannot be obtained by utilizing detection behavior. Modify the original communication topology and back to Step 1; if $C_n = \emptyset$, continue.

Step 5: Output $\mathcal{L}' = \mathcal{D}' - \mathcal{W}'$ and DM .

Consider a multi-UGV system with 11 vehicles and define vehicle 1 as the leader which can receive the information from the virtual leader. The example is depicted in Fig. 11, where observers are pointed by dotted arrows. For the dotted arrows pointing to the same observer, the vehicles connected at their starts form an agent cluster. Notice that there are two ways for information interaction, one is realized by communication topology, and the other one is fulfilled by observation. To achieve formation tracking control, it is stipulated that communication topology must contain at least one spanning tree. Then the complete Laplacian matrix which contains the information interaction between observers and vehicles observed by them is generated according to the communication topology. Therefore, the vehicles observed by an observer will change only when the communication topology switches.

The corresponding complete Laplacian matrix is represented by

$$\mathcal{L}'^{12 \times 12} = \begin{bmatrix} 0 & \mathbf{0}^{1 \times 11} \\ \mathcal{L}'_{21}^{11 \times 1} & \mathcal{L}'_{22}^{11 \times 11} \end{bmatrix}. \quad (23)$$

Based on Definition 1, the sufficient conditions for the multi-UGV system with proposed composite potential field forces to safely accomplish time-varying formation tracking is as follows.

Definition 2. Denote A_d^q as the agent cluster which is observed by vehicle q , $q \in a_{ob}$, vehicles in A_d^q except the central vehicle A_i^q are denoted as A_j^q , vehicles in the multi-UGV system except A_d^q and q are denoted as A_k^q . For any given bounded initial states, if (9), (11) hold and

$$\begin{cases} \lim_{t \rightarrow \infty} (\theta_q^w(t) - \theta_{ob}^q(t)) = 0, q \in a_{ob}, \\ rr < \overrightarrow{\|A_i^q A_f^q\|_2} < dd_1, \\ r_{rmin} < \overrightarrow{\|A_i^q q\|_2} < dd_2, q \in a_{ob}, \\ r_{rmin} < \overrightarrow{\|A_i^q A_k^q\|_2}, \end{cases} \quad (24)$$

where $\theta_{ob}^q(t)$ is the yaw command of observer q , which is defined by the relative position of vehicle q and its corresponding agent cluster. Then the multi-UGV system with proposed composite potential field forces is said to safely accomplish time-varying formation tracking. Specially, $h_i(t)$ in (9) is subject to the range of potential field and potential field forces will vanish when the desired formation structure is fulfilled.

Remark 9. Instead of $\mathcal{L}$, the complete Laplacian matrix $\mathcal{L}'$ which consists of detection and communication links is utilized. From Definition 2, one sees that the formation tracking control is divided into position and velocity control, and yaw control. Moreover, (24) is a specific expression of (10) when composite potential field forces are employed. Different from [22–27,31], Definition 2 provides all factors of APF-based formation tracking for multi-UGV system on a directed graph.

4.2. Design of control protocol

Based on Section 2.2, the following lemmas are given.

Lemma 1 ([39]). According to the previous description of $\mathcal{G}$, all eigenvalues of $\mathcal{L}'_{22}$ have positive real parts.

Lemma 2 ([40]). A diagonal matrix $Dm = \text{diag}\{dm_1, dm_2, \dots, dm_N\}$ whose diagonal elements are positive can be found to make $\psi = Dm\mathcal{L}'_{22} + \mathcal{L}'_{22}^T Dm > 0$ hold, where $[dm_1, dm_2, \dots, dm_N]^T = (\mathcal{L}'_{22})^{-1} \mathbf{1}_N$ and $\mathbf{1}_N$ stands for a column vector with N entries equal to 1.

For $i, j \in 1, 2, \dots, N$, the error of time-varying formation tracking for vehicle i is defined as

$$\xi_i(t) = \sum_{j \in 1, 2, \dots, N} \omega_{ij}((x_i(t) - h_i(t)) - (x_j(t) - h_j(t))) + \omega_{il}(x_i(t) - x_l(t) - h_i(t)),$$

where $\omega_{il} > 0$ if and only if vehicle i contains the virtual leader as its neighbor and $\omega_{il} = 0$ otherwise.

Let $\xi(t) = [\xi_1^T(t), \xi_2^T(t), \dots, \xi_N^T(t)]^T$, $x(t) = [x_1^T(t), x_2^T(t), \dots, x_N^T(t)]^T$, $\tilde{x}(t) = x(t) - \mathbf{1}_N \otimes x_l(t) - h(t)$, $\mathbf{I}_n = \mathbf{I}_m \otimes \mathbf{I}_o$, $\mathbf{I}_o$ is an identity matrix with dimension o which indicates the order of (6), it can be obtained that

$$\begin{aligned} \xi(t) &= (\mathcal{L}'_{22} \otimes \mathbf{I}_m \otimes \mathbf{I}_o)(x(t) - \mathbf{1}_N \otimes x_l(t) - h(t)) \\ &= (\mathcal{L}'_{22} \otimes \mathbf{I}_m \otimes \mathbf{I}_o)\tilde{x}(t) \\ &= \mathcal{L}'_{22} \otimes \mathbf{I}_n \tilde{x}(t). \end{aligned} \quad (25)$$

Consider the following formation tracking protocol

$$u_i(t) = C_1 \xi_i(t) + v_i(t) - \delta g_i(t) + f_i(t), i \in 1, 2, \dots, N, \quad (26)$$

where $\delta \geq \tau > 0$, $g_i(t) = \frac{\tilde{B}^T \tilde{P} \tilde{\xi}_i(t)}{\|\tilde{B}^T \tilde{P} \tilde{\xi}_i(t)\|_2}$ when $\|\tilde{B}^T \tilde{P} \tilde{\xi}_i(t)\|_2 > 0$ and $g_i(t) = \mathbf{0}$ when $\|\tilde{B}^T \tilde{P} \tilde{\xi}_i(t)\|_2 = 0$. $\tilde{P} = \mathbf{I}_m \otimes P$, P donates the positive definite solution of the algebraic Riccati equation $A^T P + PA - PBB^T P + I_o = \mathbf{0}$, and the solution exists when (A, B) is stabilizable [41]. $v_i(t)$ is compensation input of formation tracking and $\lim_{t \rightarrow \infty} (\hat{A}h_i(t) - \dot{h}_i(t) + \tilde{B}v_i(t)) = \mathbf{0}$. Besides, $C_1 = -I_m \otimes (cB^T P)$ and $c \geq \frac{\max_{\lambda}(\text{Dm})}{\min_{\lambda}(\psi)}$.

Let vehicle j be the center of an agent cluster which is observed by vehicle i , denote v_{ij} as the vector from vehicle i to vehicle j , the angle between v_{ij} and the positive direction of x_i^w is $\theta_{ij} \in [0, 2\pi)$. Then one has

$$\theta_{ij} = \begin{cases} 0, & x_j^w > x_i^w, y_j^w = y_i^w, \\ \frac{\pi}{2}, & x_j^w = x_i^w, y_j^w > y_i^w, \\ \pi, & x_j^w < x_i^w, y_j^w = y_i^w, \\ \frac{3\pi}{2}, & x_j^w = x_i^w, y_j^w < y_i^w, \\ \arctan(\frac{y_j^w - y_i^w}{x_j^w - x_i^w}), & x_j^w > x_i^w, y_j^w > y_i^w, \\ \pi + \arctan(\frac{y_j^w - y_i^w}{x_j^w - x_i^w}), & x_j^w < x_i^w, y_j^w > y_i^w, \\ \pi + \arctan(\frac{y_j^w - y_i^w}{x_j^w - x_i^w}), & x_j^w < x_i^w, y_j^w < y_i^w, \\ 2\pi + \arctan(\frac{y_j^w - y_i^w}{x_j^w - x_i^w}), & x_j^w > x_i^w, y_j^w < y_i^w, \end{cases} \quad (27)$$

$$\dot{\theta}_{ij} = \frac{(\dot{y}_j^w - \dot{y}_i^w)(x_j^w - x_i^w) - (\dot{x}_j^w - \dot{x}_i^w)(y_j^w - y_i^w)}{(x_j^w - x_i^w)^2 + (y_j^w - y_i^w)^2}, \quad (28)$$

$$\ddot{\theta}_{ij} = \frac{[(\ddot{y}_j^w - \ddot{y}_i^w)(x_j^w - x_i^w) - (\ddot{x}_j^w - \ddot{x}_i^w)(y_j^w - y_i^w)][(x_j^w - x_i^w)^2 + (y_j^w - y_i^w)^2]}{((x_j^w - x_i^w)^2 + (y_j^w - y_i^w)^2)^2} \\ - \frac{[2(x_j^w - x_i^w)(\dot{x}_j^w - \dot{x}_i^w) + 2(y_j^w - y_i^w)(\dot{y}_j^w - \dot{y}_i^w)](\dot{y}_j^w - \dot{y}_i^w)(x_j^w - x_i^w)}{((x_j^w - x_i^w)^2 + (y_j^w - y_i^w)^2)^2} \\ + \frac{[2(x_j^w - x_i^w)(\dot{x}_j^w - \dot{x}_i^w) + 2(y_j^w - y_i^w)(\dot{y}_j^w - \dot{y}_i^w)](\dot{x}_j^w - \dot{x}_i^w)(y_j^w - y_i^w)}{((x_j^w - x_i^w)^2 + (y_j^w - y_i^w)^2)^2}. \quad (29)$$

Notice that θ_{ij} and θ_i^w are discontinuous at the boundary 2π , vehicles that should be observed cannot stay within the detection region forever. To make vehicle i observe vehicle j constantly, the error of yaw control is given by

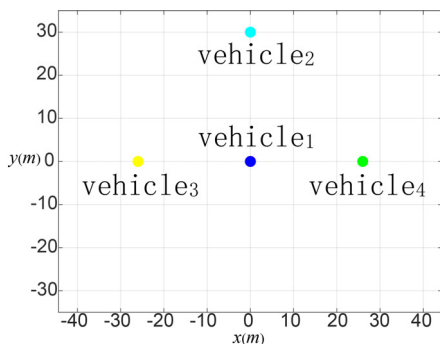
$$\eta_i = \begin{cases} \theta_i^w - \theta_{ij} + 2\pi, & \text{if } \theta_i^w < \frac{\pi}{2} \wedge \theta_{ij} > \frac{3\pi}{2}, \\ \theta_i^w - \theta_{ij} - 2\pi, & \text{if } \theta_i^w > \frac{3\pi}{2} \wedge \theta_{ij} < \frac{\pi}{2}, \\ \theta_i^w - \theta_{ij}, & \text{otherwise.} \end{cases} \quad (30)$$

Denote p , C_2 and C_3 as positive constants, the sliding surface is designed as $S = p\eta_i + \dot{\eta}_i$, then it follows that

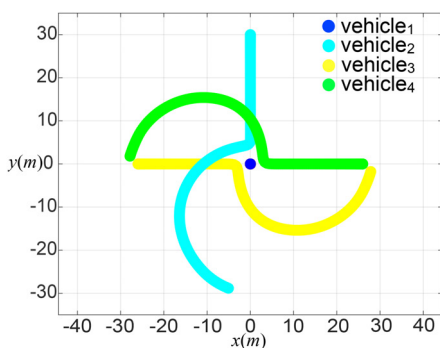
$$\begin{aligned} \dot{S} &= p\dot{\eta}_i + \ddot{\eta}_i = -C_2 \text{sign}(S) - C_3 S \\ &= p\dot{\eta}_i + u_i^b - f_i^b - \ddot{\theta}_{ij}. \end{aligned} \quad (31)$$

Yaw control protocol is given by

$$u_i^b = \begin{cases} \ddot{\theta}_{ij} + f_i^b - p\dot{\eta}_i - C_2 \text{sign}(S) - C_3 S, & \text{if } i \in a_{ob}, \\ 0, & \text{otherwise.} \end{cases} \quad (32)$$



(a) Initial state.



(b) Trajectories of vehicles when applying cross-repulsive potential field force.

Fig. 12. Initial state and trajectory of multi-UGV system with the improved repulsive potential field force.

4.3. Analysis of convergence

The compact form of (26) is given by

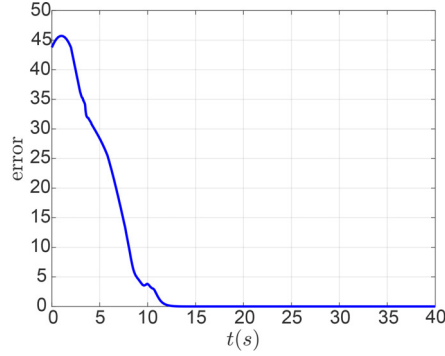
$$u(t) = I_N \otimes C_1 \xi(t) - \delta g(t) + v(t) + F(t),$$

where $u(t) = [u_1^T(t), u_2^T(t), \dots, u_N^T(t)]^T$, $v(t) = [v_1^T(t), v_2^T(t), \dots, v_N^T(t)]^T$, $F(t) = [f_1^T(t), f_2^T(t), \dots, f_N^T(t)]^T$, $g(t) = [g_1^T(t), g_2^T(t), \dots, g_N^T(t)]^T$.

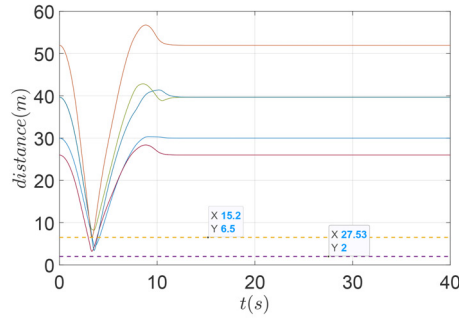
Denote $f^p = [f_1^p, f_2^p, \dots, f_N^p]$ as the resultant potential field force which is determined by $\mathbf{F}_{M1} + \mathbf{F}_{M3} + \mathbf{F}_{M2} + \mathbf{F}\mathbf{F}$. Then the whole control protocol $U(t)$ can be integrated as $[(u_1 + f_1^p)^T, u_1^b, (u_2 + f_2^p)^T, u_2^b, \dots, (u_N + f_N^p)^T, u_N^b]^T$.

By substituting protocol (26) into (8), one has

$$\begin{aligned} \dot{x} &= I_N \otimes \tilde{A}x + I_N \otimes \tilde{B}(u - F) \\ &= I_N \otimes \tilde{A}x + I_N \otimes \tilde{B}(I_N \otimes C_1 \xi + v - \delta g) \\ &= I_N \otimes \tilde{A}x + I_N \otimes (\tilde{B}C_1)\xi + I_N \otimes \tilde{B}v - \delta I_N \otimes \tilde{B}g. \end{aligned} \quad (33)$$



(a) System error based on cross-repulsive potential field force.



(b) Distance curves between vehicles based on cross-repulsive potential field force.

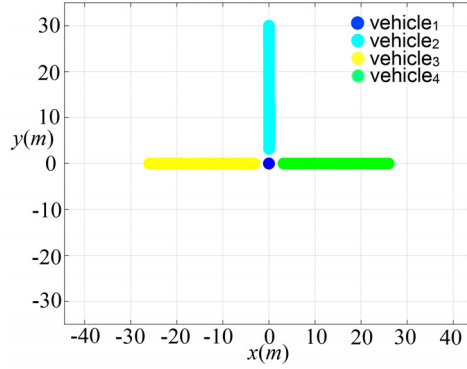
Fig. 13. Formation error and distance curves of multi-UGV system with collision avoidance.

The derivative of ξ is as follows

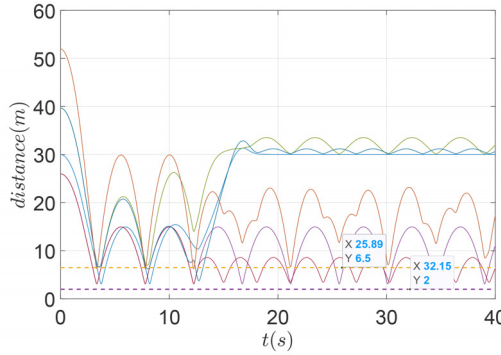
$$\begin{aligned}
 \dot{\xi} &= (\mathcal{L}'_{22} \otimes \mathbf{I}_n)(\dot{\tilde{x}}) \\
 &= (\mathcal{L}'_{22} \otimes \mathbf{I}_n)(\dot{x} - \mathbf{1}_N \otimes \dot{x}_l - \dot{h}) \\
 &= I_N \otimes \tilde{A}\xi + \mathcal{L}'_{22} \otimes \tilde{A}h \\
 &\quad + \mathcal{L}'_{22} \otimes (\tilde{B}C_1)\xi + \mathcal{L}'_{22} \otimes \tilde{B}v - \delta \mathcal{L}'_{22} \otimes \tilde{B}g \\
 &\quad - \mathcal{L}'_{22} \mathbf{1}_N \otimes (\tilde{B}u_l) - (\mathcal{L}'_{22} \otimes \mathbf{I}_n)\dot{h}.
 \end{aligned} \tag{34}$$

For the multi-UGV system with collision avoidance and detection behavior, the following results can be derived.

Theorem 2. For a multi-UGV system with N vehicles and all individuals are described by (8), based on the complete matrix which is obtained by Algorithm 1, the multi-UGV system can accomplish time-varying formation tracking with collision avoidance and detection behavior by using the whole control protocol $U(t)$.



(a) Trajectories of vehicles when applying traditional APF-based method.



(b) Distance curves between vehicles based on traditional APF-based method.

Fig. 14. Initial state and trajectory of multi-UGV system with traditional APF-based method.

Proof. According to Theorem 1, $f_i^p + u_i \equiv 0$ does not hold by virtue of the eradication of local minima and deadlock state, which means the potential field forces will not be continuously generated during the whole formation tracking process as long as the desired formation structure described by $h(t)$ is reasonable. Then $\exists t_1 > 0$, when $t \geq t_1$, $f_i^p(t) + u_i(t) \equiv u_i(t)$ and $U(t) \equiv [u_1^T, u_1^b, u_2^T, u_2^b, \dots, u_N^T, u_N^b]^T$ hold. Without local minima or deadlock dilemma, it is only necessary to analyze the convergence when $t \geq t_1$, the investigation is as follows.

Choose Lyapunov functions as $V_1 = \frac{1}{2}\xi^T(Dm \otimes \tilde{P})\xi$ and $V_2 = \frac{1}{2}S^T S$, which are positive definite apparently. Taking the derivatives of V_1 yields

$$\begin{aligned} \dot{V}_1 = & \frac{1}{2}\xi^T(Dm \otimes (\tilde{P}\tilde{A} + \tilde{A}^T\tilde{P}))\xi + \xi^T(Dm\mathcal{L}'_{22} \otimes (\tilde{P}\tilde{A}))h \\ & + \frac{1}{2}\xi^T(\psi \otimes (\tilde{P}\tilde{B}C_1))\xi + \xi^T(Dm\mathcal{L}'_{22} \otimes (\tilde{P}\tilde{B}))v - \xi^T(Dm\mathcal{L}'_{22} \otimes \tilde{P})\dot{h} \\ & - \xi^T(Dm\mathcal{L}'_{22}\mathbf{1}_N \otimes (\tilde{P}\tilde{B}u_l)) - \delta\xi^T(Dm\mathcal{L}'_{22} \otimes (\tilde{P}\tilde{B}))g. \end{aligned}$$

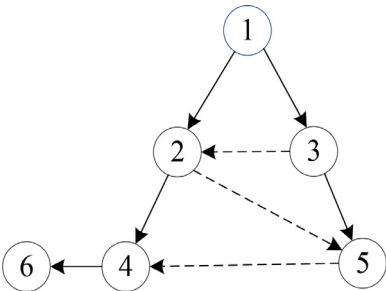
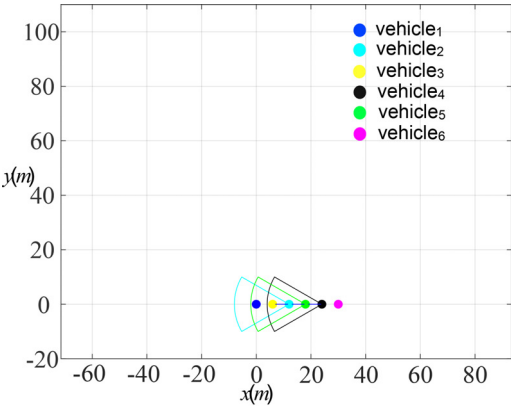
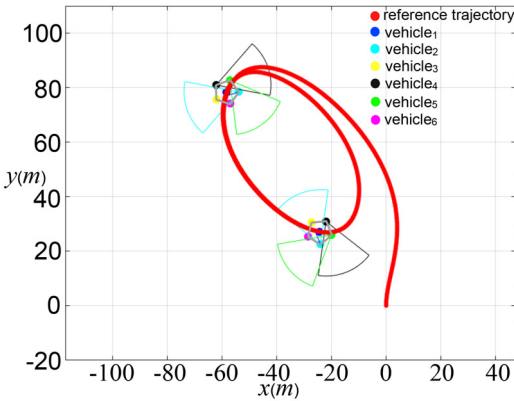


Fig. 15. Topology with detection behaviors.

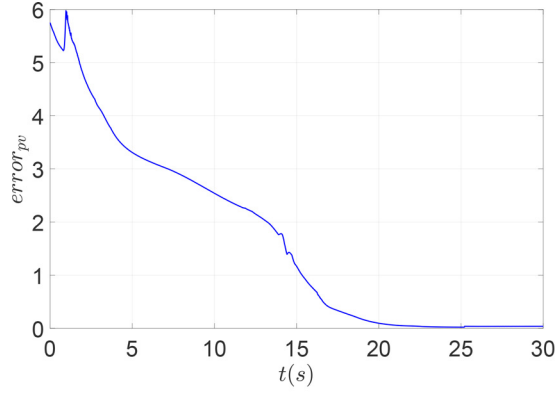


(a) Initial state.

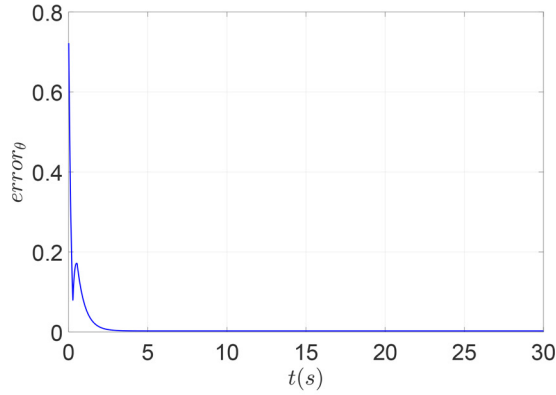


(b) Formation pattern.

Fig. 16. Initial state and trajectory of multi-UGV system with single-to-single detection behavior and collision avoidance.



(a) Error of formation tracking.

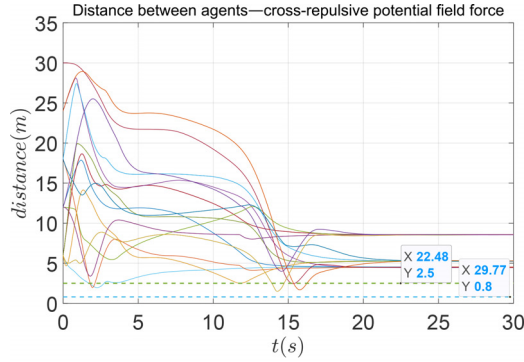


(b) Error of yaw control.

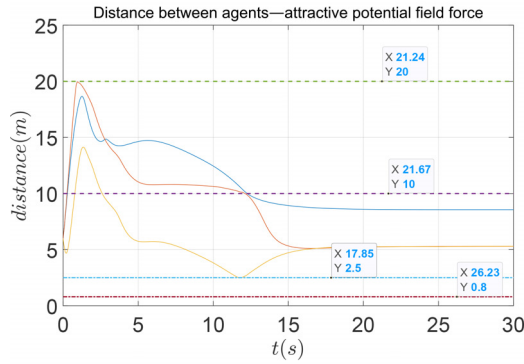
Fig. 17. Error of multi-UGV system with single-to-single detection behavior and collision avoidance.

From $c \geq \frac{\max_{\lambda}(Dm)}{\min_{\lambda}(\psi)}$, one has

$$\begin{aligned}
 \dot{V}_1 \leq & \frac{1}{2} \xi^T (Dm \otimes (\tilde{P}\tilde{A} + \tilde{A}^T \tilde{P})) \xi + \xi^T (Dm \mathcal{L}'_{22} \otimes (\tilde{P}\tilde{A})) h \\
 & - \frac{1}{2} \xi^T \left(\frac{\max_{\lambda}(Dm)}{\min_{\lambda}(\psi)} \psi \otimes (\tilde{P}\tilde{B}\tilde{B}^T \tilde{P}) \right) \xi + \xi^T (Dm \mathcal{L}'_{22} \otimes (\tilde{P}\tilde{B})) v \\
 & - \sum_{i=1}^N dm_i \omega_{il} \xi_i^T \tilde{P} \tilde{B} u_l - \xi^T (Dm \mathcal{L}'_{22} \otimes \tilde{P}) \dot{h} \\
 & - \delta \sum_{i=1}^N dm_i \xi_i^T \tilde{P} \tilde{B} \left(\sum_{j=1}^N \omega_{ij} (g_i - g_j) + \omega_{il} g_i \right).
 \end{aligned}$$



(a) Distance curves between vehicles.



(b) Distance curves between observers and vehicles

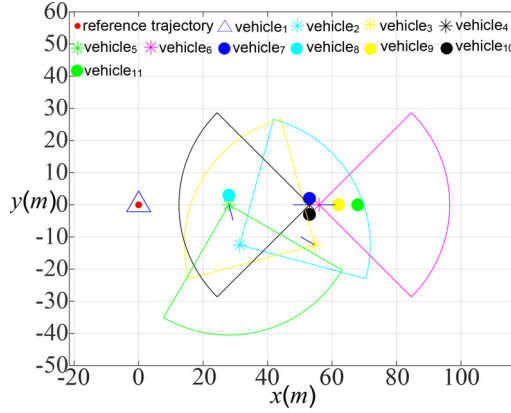
Fig. 18. Distance curves of multi-UGV system with single-to-single detection behavior and collision avoidance.

Based on $g_i(t) = \frac{\tilde{B}^T \tilde{P} \xi_i(t)}{\|\tilde{B}^T \tilde{P} \xi_i(t)\|_2}$, one can obtain

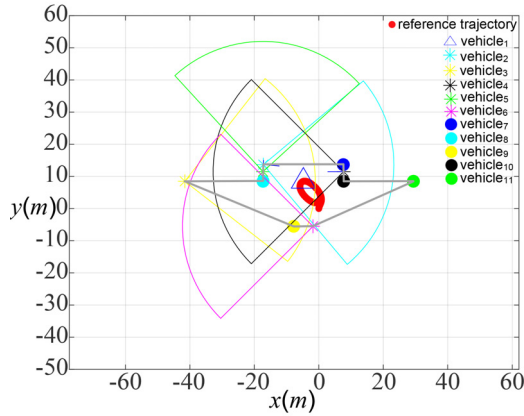
$$\begin{aligned} \dot{V}_1 \leq & \frac{1}{2} \xi^T (Dm \otimes (\tilde{P}\tilde{A} + \tilde{A}^T \tilde{P})) \xi + \xi^T (Dm \mathcal{L}'_{22} \otimes (\tilde{P}\tilde{A})) h \\ & - \frac{1}{2} \xi^T (Dm \otimes (\tilde{P}\tilde{B}\tilde{B}^T \tilde{P})) \xi + \xi^T (Dm \mathcal{L}'_{22} \otimes (\tilde{P}\tilde{B})) v \\ & - \xi^T (Dm \mathcal{L}'_{22} \otimes \tilde{P}) \dot{h} \\ & + \sum_{i=1}^N dm_i \omega_{il} \|\tilde{B}^T \tilde{P} \xi_i\|_2 \|u_l\|_2 \\ & - \delta \sum_{i=1}^N dm_i \omega_{il} \|\tilde{B}^T \tilde{P} \xi\|_2. \end{aligned}$$

According to $\delta \geq \tau > 0$, one has

$$\dot{V}_1 \leq \frac{1}{2} \xi^T (Dm \otimes (\tilde{P}\tilde{A} + \tilde{A}^T \tilde{P})) \xi + \xi^T (Dm \mathcal{L}'_{22} \otimes (\tilde{P}\tilde{A})) h$$



(a) Initial state.



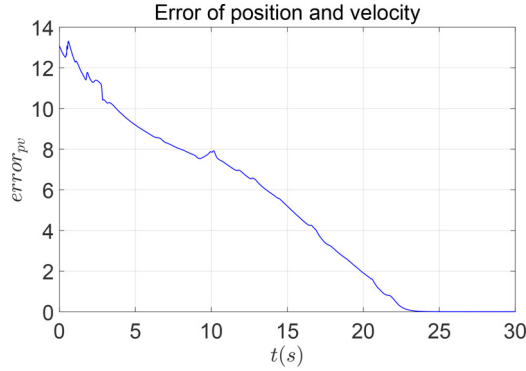
(b) Formation pattern.

Fig. 19. Initial state and trajectory of multi-UGV system with single-to-multiple detection behavior and collision avoidance.

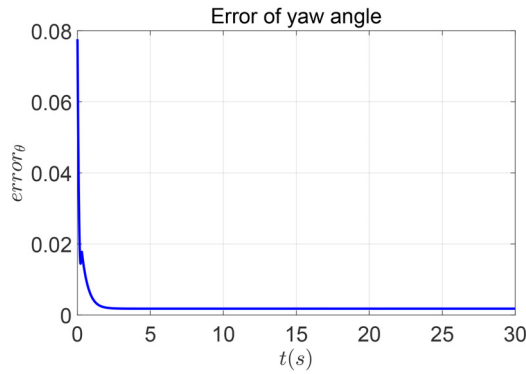
$$\begin{aligned}
 & -\frac{1}{2}\xi^T(Dm \otimes (\tilde{P}\tilde{B}\tilde{B}^T\tilde{P}))\xi + \xi^T(Dm\mathcal{L}'_{22} \otimes (\tilde{P}\tilde{B}))v \\
 & -\xi^T(Dm\mathcal{L}'_{22} \otimes \tilde{P})\dot{h} \\
 & = \frac{1}{2}\xi^T(Dm \otimes (\tilde{P}\tilde{A} + \tilde{A}^T\tilde{P} - (\tilde{P}\tilde{B}\tilde{B}^T\tilde{P})))\xi \\
 & + \xi^T(\sqrt{Dm} \otimes I_n)(\sqrt{Dm}\mathcal{L}'_{22} \otimes \tilde{P})((I_N \otimes \tilde{A})h + (I_N \otimes \tilde{B})v - \dot{h}).
 \end{aligned}$$

Inspired by Youngs inequality, when $0 < \varepsilon < 1$, one can obtain

$$\begin{aligned}
 \dot{V}_1 & \leq \frac{1}{2}\xi^T(Dm \otimes (\tilde{P}\tilde{A} + \tilde{A}^T\tilde{P} - (\tilde{P}\tilde{B}\tilde{B}^T\tilde{P}) + (1 - \varepsilon)I_n))\xi \\
 & + \frac{1}{2} \frac{1}{1 - \varepsilon} \|(\sqrt{Dm}\mathcal{L}'_{22} \otimes \tilde{P})((I_N \otimes \tilde{A})h + (I_N \otimes \tilde{B})v - \dot{h})\|^2
 \end{aligned}$$



(a) Error of formation tracking.



(b) Error of yaw control.

Fig. 20. Error of multi-UGV system with single-to-multiple detection behavior and collision avoidance.

$$\leq -\frac{1}{2} \frac{\varepsilon}{\max_{\lambda}(\tilde{P})} V_1 + \frac{1}{2} \frac{1}{1-\varepsilon} \|(\sqrt{Dm} \mathcal{L}'_{22} \otimes \tilde{P})((I_N \otimes \tilde{A})h + (I_N \otimes \tilde{B})v - \dot{h})\|^2.$$

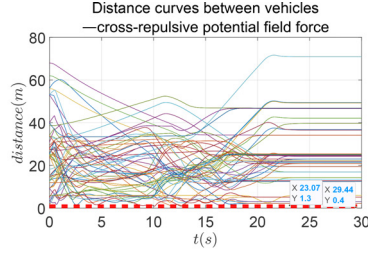
Since $\lim_{t \rightarrow \infty} (\tilde{A}h_i(t) - \dot{h}_i(t) + \tilde{B}v_i(t)) = \mathbf{0}$, according to Lemma 2.19 in [42], the system is asymptotically convergent and $\lim_{t \rightarrow \infty} V_1 = 0$, i.e., $\lim_{t \rightarrow \infty} \xi = 0$.

Taking the derivatives of V_2 yields

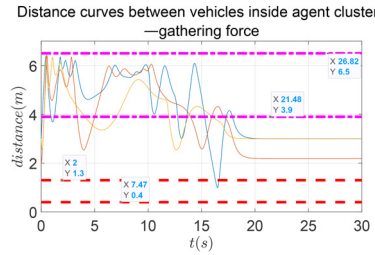
$$\begin{aligned} \dot{V}_2 &= S^T \dot{S} \\ &= S^T [p\dot{\eta}_i + u_i^b - f_i^b - \ddot{\theta}_{ij}], i \in a_{ob}. \end{aligned} \quad (35)$$

Substituting (32) into (35) leads to

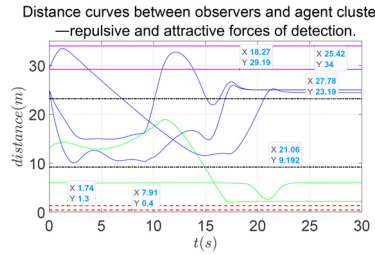
$$\begin{aligned} \dot{V}_2 &= -C_2 S^T \text{sign}(S) - C_3 S^T S \\ &= -C_2 \|S\|_1 - C_3 S^T S \\ &\leq 0. \end{aligned} \quad (36)$$



(a) Distance curves between vehicles.



(b) Distance curves between vehicles inside the agent clusters.



(c) Distance curves between observers and agent clusters.

Fig. 21. Distance curves of multi-UGV system with single-to-multiple detection behavior and collision avoidance.

Thus, $\lim_{t \rightarrow \infty} \xi = 0$, $\lim_{t \rightarrow \infty} \eta_i = 0$ and $\lim_{t \rightarrow \infty} \dot{\eta}_i = 0$ hold simultaneously, i.e., (9) and the first function of (24) are satisfied. According to Theorem 1, (11) and the rest functions of (24) can be ensured. Then, a multi-UGV system on directed topology which is not unilaterally connected can accomplish time-varying formation tracking by protocol (26), moreover, collision avoidance and detection behavior are fulfilled by protocol (32) and resultant potential field force f^p . This completes the proof of Theorem 2. $\square$

Remark 10. To avoid the large chattering, $g_i(t)$ and $\text{sign}(S)$ can be fitted by a continuous function, e.g., $g_i(t)$ can be replaced by $g'_i(t) = \frac{\bar{B}^T \bar{P} \xi_i(t)}{\|\bar{B}^T \bar{P} \xi_i(t)\|_2 + \chi}$, where χ is a small positive constant and $\xi(t)$ will asymptotically converge to a bounded set whose norm is related to χ [34].

5. Numerical simulations

To validate the effectiveness of the proposed methods, simulations are conducted for collision avoidance and single-to-multiple detection based on the model of UGV.

5.1. Collision avoidance

Consider a multi-UGV system composed of four vehicles, where a leader is fixed at the origin and three followers need to reach the other sides of the origin. Let $r_2 = 6.5$, $rr = 2$. Fig. 12 gives the initial state and trajectory of each vehicle. Based on the cross-repulsive potential field force, each follower circumnavigates around the central leader and arrives at the destination without collision.

Fig. 13(a) indicates that the system error converges to 0. Moreover, cross-repulsive potential field force is activated and there is no collision between vehicles as shown in Fig. 13(b).

For comparison, the traditional APF whose repulsive force is generated along the connecting line between two individuals is applied and the results are shown in Fig. 14. Take a follower in Fig. 14(a) as an example for analysis. The potential field force increases when the vehicle moves toward the central individual and the effect of the formation control protocol will be weakened to zero. Then the individual will be bounced away and the effect of the formation control protocol is gradually restored. The vehicle will decelerate until it stops and move towards the central individual again. The above process is repeated and the deadlock state occurs. The oscillation of the curves in Fig. 14(b) can also demonstrate the appearance of local minima.

5.2. Formation tracking with single-to-single detection

Consider a multi-UGV system composed of six vehicles, where a leader tracks the reference trajectory $[5\cos(0.1\pi t + \pi/3), 5\sin(0.1\pi t)]^T$, and the desired formation vector $h = [h_1^T, h_2^T, \dots, h_6^T]^T$ is described by $h = [0, 0, 0, 0, 4.5\cos(0.1t + \frac{9\pi}{10}), -0.45\sin(0.1t + \frac{9\pi}{10}), 4.5\sin(0.1t + \frac{9\pi}{10}), 0.45\cos(0.1t + \frac{9\pi}{10}), 4.5\cos(0.1t + \frac{\pi}{10}), -0.45\sin(0.1t + \frac{\pi}{10}), 4.5\sin(0.1t + \frac{\pi}{10}), 0.45\cos(0.1t + \frac{\pi}{10}), 4.5\cos(0.1t + \frac{17\pi}{10}), -0.45\sin(0.1t + \frac{17\pi}{10}), 4.5\sin(0.1t + \frac{17\pi}{10}), 0.45\cos(0.1t + \frac{17\pi}{10}), 4.5\cos(0.1t + \frac{13\pi}{10}), -0.45\sin(0.1t + \frac{13\pi}{10}), 4.5\sin(0.1t + \frac{13\pi}{10}), 0.45\cos(0.1t + \frac{13\pi}{10}), 4.5\cos(0.1t + \frac{\pi}{2}), -0.45\sin(0.1t + \frac{\pi}{2}), 4.5\sin(0.1t + \frac{\pi}{2}), 0.45\cos(0.1t + \frac{\pi}{2})]^T$. Let $rr = 0.8$, $r_2 = 2.5$, $r_3 = 10$, $d = 20$, $c = 6/1.3329$, $\delta = 7$. The topology between vehicles is depicted in Fig. 15, where dotted lines represent detection behaviors and vehicle 1 is the actual leader. The initial state and trajectory of time-varying formation tracking are shown in Fig. 16, where vehicles 1–6 are represented by the blue, cyan, yellow, black, green, and magenta dots, three sectors are detection ranges of different observers, each angular bisector denoted by the blue line represents the orientation of corresponding observer. From Fig. 16(b), the actual leader can track the reference trajectory marked by the red curve, and the followers can form the desired time-varying formation structure.

From Fig. 17(a) and Fig. 17(b), the effectiveness of control protocols (26) and (32) is validated.

According to Fig. 18(a) and Fig. 18(b), the APF-based method which is designed for single-to-single detection is activated, and the distances between different vehicles are within the allowable range. Therefore, the composite potential field force is effective.

5.3. Formation tracking with single-to-multiple detection

Consider a multi-UGV system composed of eleven vehicles, where vehicle 1 tracks the reference trajectory $[2\cos(0.1\pi t + \pi/3), 2\sin(0.1\pi t)]^T$, the desired formation vector $h = [h_1^T, h_2^T, \dots, h_{11}^T]^T$ is described by $[0, 0, 0, 0, -12.5 + (25 - 25\cos(\pi/36)), 0, 25\sin(\pi/36) + 3, 0, -(25\cos(\arcsin(3/12.5)) + 12.5), 0, 0, 0, 12.5, 0, 3, 0, -12.5, 0, 3, 0, 3, 0, -14, 0, 12.5 - (25 - 25\cos(\pi/36)), 0, 25\sin(\pi/36) + 3, 0, -12.5, 0, 0, -3, 0, -14, 0, 12.5, 0, 0, 0, (25\cos(\pi/6) + 12.5), 0, 0, 0]^T$. Let $rr = 0.4$, $r_2 = 1.3$, $r_3 = 3.9$, $dd_1 = 6.5$, $r_{min} \approx 9.1942$, $r_{max} \approx 23.1942$, $d = 40.5$, $dd_2 = 34$, $dd_3 = 29.2$, $c = 11/1.2751$, $\delta = 7$. The topology of the system with detection behaviors is depicted as Fig. 11(b), where dotted lines represent detection behaviors and vehicle 1 is the leader. The initial state and trajectory of the system are shown in Fig. 19, where vehicle 1 is marked by a blue triangle, vehicles 2–6 are represented by cyan, yellow, black, green, magenta asterisks, vehicles 7–11 are denoted by blue, cyan, yellow, black, green dots. Five sectors are detection areas of different observers, each angular bisector represented by the blue line is the orientation of the corresponding observer. From Fig. 19(b), the leader can track the reference trajectory represented by the red curve, and the followers can form the desired formation structure.

From Fig. 20(a) and Fig. 20(b), the effectiveness of control protocols (26) and (32) is validated.

Based on Fig. 21(a), Fig. 21(b) and Fig. 21(c), the composite potential field force which is designed for single-to-multiple detection is activated and various distances between different vehicles are within their allowable ranges respectively. Therefore, the validity of the composite potential field force is verified.

6. Conclusion

Time-varying formation tracking for the multi-UGV system with collision avoidance on directed topology was investigated in this paper. Local minima or deadlock dilemma was eradicated by the improved repulsive potential field force. Detection behavior and its generation algorithm were introduced to prevent the failure of collision avoidance from local information in the directed graph which is not unilaterally connected. Moreover, the proposed composite potential field forces and control protocols can make the multi-UGV system achieve formation tracking safely while ensuring the observers continuously detect their agent clusters.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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